



CEBU INSTITUTE OF TECHNOLOGY
U N I V E R S I T Y

IT342-G1

System Integration and Architecture

System Design Document (SDD)

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Prepared By: Cruz, John Harley Abellana

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REVISION HISTORY TABLE

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0.1	2/21/2026	John Harley A. Cruz	Initial draft	Draft
1.0	2/28/2026	John Harley A. Cruz	API Contracts / Communication → API Specifications & Interaction Database Design → Database Structure Planning UI/UX → User Interface & User Experience Design Plan → Project Strategy	Done
1.1	3/7/2026	John Harley A. Cruz	Updated History Table	Done
0.4	[Date]	[Your Name]	Added UI/UX designs	Review
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EXECUTIVE SUMMARY

1.1 Project Overview

RateFlow MVP is a simplified service-based rating and feedback management platform that allows users to submit ratings and reviews for services while enabling administrators to monitor and manage feedback efficiently.

1.2 Objectives

1. Develop a fully functional service rating MVP with authentication and feedback submission
2. Implement a **three-tier architecture** using Spring Boot (backend), React (web), and Android (mobile)
3. Create RESTful APIs for communication between frontend and backend systems
4. Allow users to submit, edit, and view ratings and feedback
5. Enable administrators to monitor and manage submitted feedback
6. Design a responsive and user-friendly interface for both web and mobile platforms
7. Deploy system components in a production-ready environment

1.3 Scope

Included Features:

- User registration and authentication (email/password)

- User profile management
- Submit ratings (e.g., 1–5 stars) with written feedback
- View submitted ratings and feedback history
- Admin dashboard for monitoring and managing feedbacks
- Search and filter feedback entries
- RESTful API integration
- Relational database (MySQL)
- Responsive React web interface
- Native Android mobile application
- Forgot Password System

Excluded Features:

- Social media login integration
- Real-time push notifications
- Advanced analytics dashboard with AI insights
- Payment system integration
- Automated moderation using AI
- Multi-language support
- Third-party API integrations

1.0 INTRODUCTION

1.1 Purpose

This document serves as the comprehensive design specification for the **RateFlow MVP** system. It provides detailed functional and non-functional requirements, system architecture, API specifications, database design, and implementation guidelines.

The purpose of this document is to:

- Clearly define the overall system structure and component interactions

- Establish the architectural framework (frontend, backend, and database layers)
- Specify RESTful API contracts for seamless communication between web, mobile, and backend services
- Outline the database schema and data management strategy
- Provide a development roadmap to guide implementation and deployment

This document ensures that all system components are aligned, integrated properly, and developed according to a consistent technical standard, minimizing integration issues and improving overall system reliability.

2.0 FUNCTIONAL REQUIREMENTS SPECIFICATION

Project Name: RateFlow MVP

Domain: Service Rating & Feedback Management

Primary Users:

- Customers / End-Users
- Administrators

Problem Statement:

Organizations and service providers need a simple and efficient way to collect, manage, and analyze customer feedback without relying on complex or expensive enterprise feedback systems.

Solution:

A minimalist rating and feedback platform that allows users to submit service ratings and written reviews, while administrators can monitor, manage, and analyze feedback across web and mobile platforms with a consistent user experience.

2.2 Core User Journeys

Journey 1: First-time User Submits Feedback

1. User visits web or mobile application

2. Clicks "**Sign Up**" and creates an account
 3. Logs in successfully
 4. Selects a service to rate
 5. Submits star rating (1–5 stars)
 6. Writes optional feedback comment
 7. Submits feedback
 8. Receives confirmation message
-

Journey 2: Returning User Submits and Views Feedback

1. User logs in with existing credentials
 2. Navigates to service list
 3. Searches for a specific service
 4. Submits a new rating and comment
 5. Views previously submitted feedback
 6. Edits or deletes own feedback
-

Journey 3: Administrator Service and Feedback Management (Ownership-Based)

1. Logs in using admin credentials.
2. Navigates to the **Admin Dashboard**.
3. Open the **Create Event Screen**.
4. **Creates a new service** to be rated by end users.

5. **Views all services they have created.**
 6. **Updates/edit details** of their existing services
 7. **Deletes services** they previously created.
 8. Selects a specific service from the Services Screen.
 9. Navigates to the **Feedback List Screen** for the selected service.
 10. **Views feedback and ratings submitted by end users** for that service.
 11. **Filters feedback** by date or rating.
 12. **Deletes inappropriate or irrelevant feedback** from end users.
-

2.3 Feature List (MoSCoW)

MUST HAVE

- User authentication (register, login, logout)
 - Submit rating (1–5 stars)
 - Submit written feedback
 - Date Created of services
 - View feedback history
 - Admin panel
 - Role-based access control
 - Forgot Password
 - Email Notification System
 - User profile section
 - Admin profile section
-

SHOULD HAVE

- Search and filter services
- Edit/delete feedback
- Basic rating statistics (average rating per service)

- Responsive design for web and mobile
-

COULD HAVE

- Service categories
 - Basic analytics dashboard
 - Feedback sorting (highest/lowest rating)
 - Notifications
-

WON'T HAVE

- Social login integration
 - AI-based sentiment analysis
 - Multi-language support
 - Third-party API integrations
-

2.4 Detailed Feature Specifications

Feature: User Authentication

Screens: Registration, Login

Fields: Username, Email, Password, Confirm Password

Validation:

- Valid email format
- Password strength requirement
- Unique email check

API Endpoints:

- **POST /auth/register**

- `POST /auth/login`
- `POST /auth/logout`

Security:

- JWT tokens
 - Password hashing using bcrypt
 - Role-based authorization (User/Admin)
-

Feature: Admin Authentication

Screens:

- Admin Login

Fields:

- Username (fixed: admin)
- Admin Email (fixed by owner gmail)
- Password

Validation:

- Username must match the predefined admin username ("admin").
- Email must match the predefined secure admin email.
- Password must match the predefined secure admin password.
- Display error message for invalid credentials.

API Endpoints:

`POST /auth/admin/login`
`POST /auth/logout`

Feature: Service Rating & Feedback

Screens: Service Listing, Service Detail

Display:

- Service name
- Description
- Average rating
- Star rating component
- Date Created

Functions:

- Submit rating (1–5 stars)
- Add written feedback
- Edit/delete own feedback

API Endpoints:

- `GET /services`
 - `GET /services/{id}`
 - `POST /ratings`
 - `PUT /ratings/{id}`
 - `DELETE /ratings/{id}`
-

Feature: Admin Service Management

Screens:

- Admin Dashboard
- Create Service Screen
- Access Control Screen

- Manage Service Screen
-

Fields (Create / Edit Service):

- Service Name
 - Description
 - Category
-

Functions:

- Create a new service to be rated
 - View list of services created by the logged-in admin
 - Edit service details of created services
 - Delete services
 - View basic rating statistics for each created service
-

Validation:

- Service name is required
 - Service name must be unique per admin
 - Only authenticated admins can create services
-

API Endpoints:

POST /admin/services
GET /admin/services
GET /admin/services/{serviceId}
PUT /admin/services/{serviceId}
DELETE /admin/services/{serviceId}

2.5 Acceptance Criteria

◆ USER ACCEPTANCE CRITERIA

AC-1: Successful User Registration

Given I am a new user

When I enter a valid email and strong password

And confirm the password correctly

And submit the registration form

Then my account should be created

And I should be able to log in successfully

AC-2: Successful Rating Submission

Given I am logged in as a user

When I select a service

And submit a rating (1–5 stars)

And optionally write feedback

And click submit

Then the rating should be saved

And it should appear in the service's feedback list

And the service's average rating should update

AC-3: User Can Only Modify Own Feedback

Given I am logged in as a user

When I attempt to edit or delete a rating I submitted

Then the system should allow the modification

But When I attempt to edit or delete another user's rating

Then the system should deny access

◆ ADMIN ACCEPTANCE CRITERIA

AC-4: Admin Creates and Manages Own Service

Given I am logged in as an administrator
When I create a new service
Then it should appear in my Services Management Screen

AC-5: Admin Edits a Service

Given I am logged in as the administrator
And a service already exists
When I update the service details
Then the changes should be saved successfully
And the updated information should be displayed

AC-6: Admin Deletes a Service

Given I am logged in as the administrator
And a service exists
When I delete the service
Then the service should be permanently removed
And it should no longer appear in the Services Management Screen

AC-7: Only Authenticated Admin Can Grant Admin Access

Given I am logged in as an administrator
When I access the Access Control screen
And I grant admin access to a user
Then the user's role should be updated to "Admin"
And the user should be able to access admin features

3.0 NON-FUNCTIONAL REQUIREMENTS

3.1 Performance Requirements

- API response time: ≤ 2 seconds for 95% of requests
- Web page load time: ≤ 3 seconds on broadband

- Mobile app cold start: ≤ 3 seconds
- Support 100 concurrent users
- Database queries complete within 500ms

3.2 Security Requirements

- HTTPS for all communications
- JWT token authentication
- Password hashing with bcrypt (salt rounds = 12)
- Admin endpoints require role verification

3.3 Compatibility Requirements

- **Web Browsers:** Chrome, Firefox, Safari, Edge (latest 2 versions)
- **Android:** API Level 24+ (Android 7.0+)
- **Screen Sizes:** Mobile (360px+), Tablet (768px+), Desktop (1024px+)
- **Operating Systems:** Windows 10+, macOS 10.15+, Linux Ubuntu 20.04+

3.4 Usability Requirements

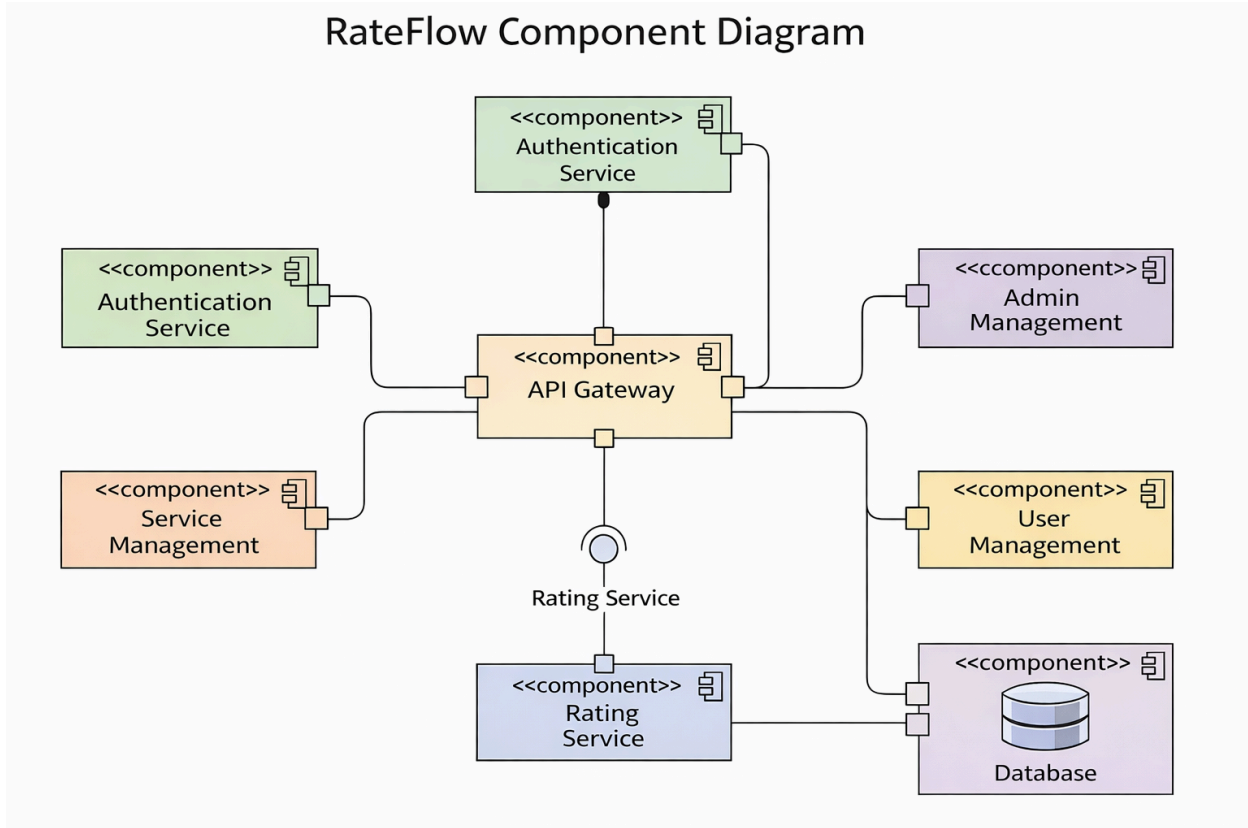
- A user shall be able to submit feedback within 5 minutes
- Admin shall be able to create a new service within 5 minutes
- Interface shall have consistent navigation across all pages
- Clear validation and error messages shall be displayed
- Forms shall prevent invalid input before submission
- Touch targets shall be at least 44x44px on mobile
- The system shall provide simple and intuitive service management for the admin

4.0 SYSTEM ARCHITECTURE

4.1 Component Diagram

Note: This should be a component diagram

RateFlow Component Diagram



Technology Stack:

- **Backend:** Spring Boot DevTools, Spring Web, Spring Data JPA, MySql Driver
- **Database:** PostgreSQL 14+
- **Web Frontend:** React 18
- **Mobile:** Kotlin
- **Build Tools:** Maven (Backend), npm/yarn (Web), Gradle (Android)
- **Deployment:** APK (Mobile) and Render (Web and Backend)
- **Security:** Spring Security, JWT authentication
- **Version Control:** Git/Github

5.0 API CONTRACT & COMMUNICATION

5.1 API Standards

Base URL	https://[server_hostname]:[port]/api/v1
Format	JSON for all requests/responses
Authentication	Bearer token (JWT) in Authorization header
Response Structure	{ "success": boolean,

	<pre> "data": object null, "error": { "code": string, "message": string, "details": object null }, "timestamp": string </pre>
--	---

5.2 Endpoint Specifications

Authentication Endpoints

User Registration

Description	User Registration
API URL	/auth/register
HTTP Request Method	POST
Format	JSON for all requests/responses
Authentication	None
Request Payload	<pre> { "username": <username>, "email": <email>, "password": <password>, "Confirmpassword": <confirmpassword> } </pre>
Response Structure	<pre> { "success": boolean, "data": { user { "Id": <id> "username": <username>, "email": <email>, }, "accessToken": <token>, "refreshToken": <token>, }, "error": { "code": string, "message": string, "details": object null }, "timestamp": string } </pre>

User Login

Description	User Login
API URL	/auth/login
HTTP Method	POST
Format	JSON for all requests/responses
Authentication	None
Request Payload	<pre>{ "username": <username> "email": <email>, "password": <password>, }</pre>
Response Structure	<pre>{ "success": boolean, "data": { "user": { "username": <username> "email": <email>, "role": <role>, }, "accessToken": <token>, "refreshToken": <token>, }, "error": { "code": string, "message": string, "details": object null }, "timestamp": string }</pre>

User Logout

Description	User Logout
API URL	/auth/login
HTTP Method	POST
Format	JSON for all requests/responses
Authentication	Bearer Token (Access Token Required)
Request Payload	<pre>{ "refreshToken": "<refreshToken>" }</pre>
Response Structure	<pre>{ "success": boolean, "data": {</pre>

	<pre> "message": "User logged out successfully" }, "error": { "code": "string", "message": "string", "details": object null }, "timestamp": "string" } </pre>
--	---

Admin Login

Description	Admin Login
API URL	/auth/admin/login
HTTP Request Method	POST
Format	JSON for all requests/responses
Authentication	None
Request Payload	<pre> { "username": <username>, "email": <email>, "password": <password>, } </pre>
Response Structure	<pre> { "success": boolean, "data": { admin{ "id": <id> "username": <username>, "email": <email>, "role": <role>, }, "accessToken": <token>, "refreshToken": <token>, }, "error": { "code": string, "message": string, "details": object null }, "timestamp": string } </pre>

Admin Logout

Description	Admin Logout
API URL	/auth/admin/login
HTTP Request Method	POST
Format	JSON for all requests/responses
Authentication	Bearer Token (Admin Access Token Required)
Request Payload	<pre>{ "refreshToken": "<refreshToken>" }</pre>
Response Structure	<pre>{ "success": boolean, "data": { "message": "Admin logged out successfully" }, "error": { "code": string, "message": string, "details": object null }, "timestamp": string }</pre>

Get All Services

Description	Retrieve list of all available services
API URL	/services
HTTP Request Method	GET
Format	JSON for all requests/responses
Authentication	Optional / Required (based on your design)
Request Payload	None
Response Structure	<pre>{ "success": boolean, "data": [{ "id": <UserIdNumber>, "serviceName": <ServiceName>, "serviceCategory": <ServiceCategory>, "serviceDescription": <Description>, }] }</pre>

	<pre> "serviceImageUrl": <serviceImageUrl>, "starRating": <starRating>, "totalStarRating": <totalStarRating>, "dateCreated": <dateCreated> }], "error": null, "timestamp": "string" } </pre>
--	---

Get Service By ID

Description	Retrieve detailed information of a specific service
API URL	/services/{id}
HTTP Request Method	GET
Format	JSON for all requests/responses
Authentication	Optional / Required (based on your design)
Request Payload	None
Response Structure	<pre> { "success": <boolean>, "data": { "id": <ServiceId>, "serviceName": <ServiceName>, "serviceCategory": <ServiceCategory>, "serviceDescription": <ServiceDescription>, "serviceImageUrl": <ServiceImageUrl>, "starRating": <StarRating>, "ratings": [{ "ratingId": <RatingId>, "userId": <UserId>, "starRating": <StarRate>, "totalStarRating": <totalStarRating>, "feedbackText": <FeedbackText>, "dateCreated": <DateCreated> }] }, "error": <null ErrorObject>, "timestamp": <Timestamp> } </pre>

	}
--	---

Submit Rating

Description	Submitting a rating to a created service
API URL	/ratings
HTTP Request Method	POST
Format	JSON for all requests/responses
Authentication	Bearer Token Required
Request Payload	<pre>{ "serviceId": <ServiceId>, "starRate": <StarRate>, "feedbackText": <FeedbackText> }</pre>
Response Structure	<pre>{ "success": <boolean>, "data": { "ratingId": <RatingId>, "serviceId": <ServiceId>, "userId": <UserId>, "starRate": <StarRate>, "feedbackText": <FeedbackText>, "dateCreated": <DateCreated> }, "error": <null ErrorObject>, "timestamp": <Timestamp> }</pre>

Update Rating

Description	Updating a rating to a created service
API URL	/ratings/{id}
HTTP Request Method	PUT
Format	JSON for all requests/responses
Authentication	Bearer Token Required
Request Payload	{

	<pre>"starRate": <StarRate>, "feedbackText": <FeedbackText> }</pre>
Response Structure	<pre>{ "success": <boolean>, "data": { "ratingId": <RatingId>, "serviceId": <ServiceId>, "userId": <UserId>, "starRate": <StarRate>, "totalStarRating": <totalStarRating>, "feedbackText": <FeedbackText>, "dateUpdated": <DateUpdated> }, "error": <null ErrorObject>, "timestamp": <Timestamp> }</pre>

Delete Rating

Description	Deleting a rating to a created service
API URL	/ratings/{id}
HTTP Request Method	DELETE
Format	JSON for all requests/responses
Authentication	Bearer Token Required
Request Payload	<pre>{ "starRate": <StarRate>, "feedbackText": <FeedbackText> }</pre>
Response Structure	<pre>{ "success": <boolean>, "data": {</pre>

	<pre> "message": <Message> }, "error": <null ErrorObject>, "timestamp": <Timestamp> } </pre>
--	--

Create Service

Description	Create a new service to be rated
API URL	/admin/services
HTTP Request Method	POST
Format	JSON for all requests/responses
Authentication	Bearer Token (Admin Only)
Request Payload	<pre> { "serviceName": <ServiceName>, "serviceCategory": <ServiceCategory>, "serviceDescription": <ServiceDescription>, "serviceImageUrl": <ServiceImageUrl> } </pre>
Response Structure	<pre> { "success": <boolean>, "data": { "serviceld": <Serviceld>, "serviceName": <ServiceName>, "serviceCategory": <ServiceCategory>, "serviceDescription": <ServiceDescription>, "serviceImageUrl": <ServiceImageUrl>, "createdBy": <CreatedBy>, "dateCreated": <DateCreated> }, "error": <null ErrorObject>, "timestamp": <Timestamp> } </pre>

--	--

Get All Services (Admin View)

Description	Retrieves all services available in the system for administrative management and review.
API URL	/admin/services
HTTP Request Method	GET
Format	JSON for all requests/responses
Authentication	Bearer Token (Admin Only)
Request Payload	<pre>{ "success": <boolean>, "data": [{ "serviceld": <Serviceld>, "serviceName": <ServiceName>, "serviceCategory": <ServiceCategory>, "serviceDescription": <ServiceDescription>, "serviceImageUrl": <ServiceImageUrl>, "starRating": <StarRating>, "createdBy": <CreatedBy>, "dateCreated": <DateCreated> }], "error": <null ErrorObject>, "timestamp": <Timestamp> }</pre>
Response Structure	<pre>{ "success": <boolean>, "data": [{ "serviceld": <Serviceld>,</pre>

	<pre> "serviceName": <ServiceName>, "serviceCategory": <ServiceCategory>, "serviceDescription": <ServiceDescription>, "serviceImageUrl": <ServiceImageUrl>, "starRating": <StarRating>, "createdBy": <CreatedBy>, "dateCreated": <DateCreated> }], "error": <null ErrorObject>, "timestamp": <Timestamp> } </pre>
--	--

Get Service By ID (Admin)

Description	Retrieves detailed information of a specific service for administrative management.
API URL	/admin/services/{serviceId}
HTTP Request Method	GET
Format	JSON for all requests/responses
Authentication	Bearer Token (Admin Only)
Request Payload	None
Response Structure	<pre> { "success": <boolean>, "data": { "serviceId": <ServiceId>, "serviceName": <ServiceName>, "serviceCategory": <ServiceCategory>, "serviceDescription": <ServiceDescription>, "serviceImageUrl": <ServiceImageUrl>, "starRating": <StarRating>, "createdBy": <CreatedBy>, </pre>

	<pre> "dateCreated": <DateCreated> }, "error": <null ErrorObject>, "timestamp": <Timestamp> } </pre>
--	--

Update Service(Admin)

Description	Allows an authenticated administrator to update the information of a specific service, including its name, category, description, and associated image.
API URL	/admin/services/{serviceId}
HTTP Request Method	PUT
Format	JSON for all requests/responses
Authentication	Bearer Token (Admin Only)
Request Payload	<pre> { "serviceName": <ServiceName>, "serviceCategory": <ServiceCategory>, "serviceDescription": <ServiceDescription>, "serviceImageUrl": <ServiceImageUrl> } </pre>
Response Structure	<pre> { "success": <boolean>, "data": { "message": <Message> }, "error": <null ErrorObject>, "timestamp": <Timestamp> } </pre>

--	--

Delete Service (Admin)

Description	Deletes a specific service from the system. This endpoint allows an authenticated administrator to permanently remove a service and its associated data.
API URL	/admin/services/{serviceId}
HTTP Request Method	DELETE
Format	JSON for all requests/responses
Authentication	Bearer Token (Admin Only)
Request Payload	None
Response Structure	<pre>{ "success": <boolean>, "data": { "message": <Message> }, "error": <null ErrorObject>, "timestamp": <Timestamp> }</pre>

5.3 Error Handling

HTTP Status Codes

- 200 OK - Successful request
- 201 Created - Resource created
- 400 Bad Request - Invalid input
- 401 Unauthorized - Authentication required/failed
- 403 Forbidden - Insufficient permissions
- 404 Not Found - Resource doesn't exist
- 409 Conflict - Duplicate resource
- 500 Internal Server Error - Server error

Error Response Format

All API errors follow a consistent response structure:

	<pre>{ "success": false, "data": null, "error": { "code": "string", "message": "string", "details": "string object null" }, "timestamp": "string" }</pre>
--	---

Error Code Examples

Example 1	<pre>json { "success": false, "data": null,</pre>
-----------	---

	<pre> "error": { "code": "AUTH-001", "message": "Invalid credentials", "details": "Email or password is incorrect" }, "timestamp": "2024-01-28T10:30:00Z" } </pre>
Example 2	<pre> { "success": false, "data": null, "error": { "code": "VALID-001", "message": "Validation failed", "details": { "email": "Email is required", "password": "Must be at least 8 characters" } }, "timestamp": "2024-01-28T10:30:00Z" } </pre>

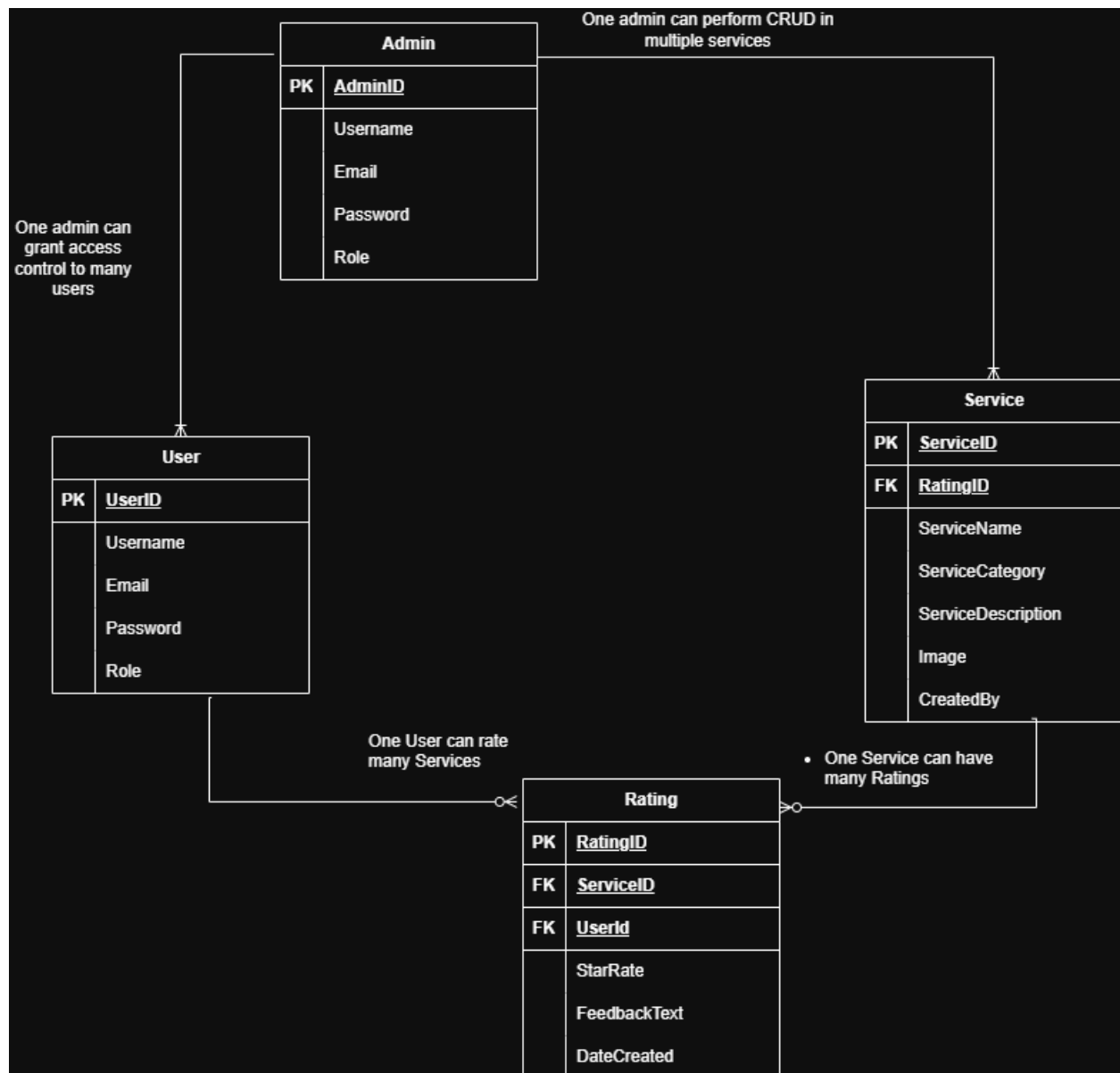
Common Error Codes

- AUTH-001: Invalid credentials
- AUTH-002: Token expired
- AUTH-003: Insufficient permissions
- VALID-001: Validation failed
- DB-001: Resource not found
- DB-002: Duplicate entry
- BUSINESS-001: User already rated this service
- SYSTEM-001: Internal server error

6.0 DATABASE DESIGN

6.1 Entity Relationship Diagram

Note: This should be an ERD (Database Schema)



Detailed Relationships:

- **One-to-Many: User → Service**

One Admin (User with Role = Admin) can create multiple Services.

Each Service is created by exactly one Admin.

- **One-to-Many: User → Rating**

One User can submit multiple Ratings.

Each Rating belongs to exactly one User.

- **One-to-Many: Service → Rating**

One Service can receive multiple Ratings.

Each Rating belongs to exactly one Service.

- **Many-to-One: Rating → User**
Many Ratings can be submitted by one User.
- **Many-to-One: Rating → Service**
Many Ratings can belong to one Service.

Key Tables:

1. **users** – User accounts and role management (User/Admin)
2. **services** – Services that can be rated
3. **ratings** – User-submitted ratings and feedback

Table Structure Summary:

users

- userID (PK)
 - username
 - email
 - password
 - role
-

services

- serviceID (PK)
 - serviceName
 - serviceCategory
 - serviceDescription
 - serviceImage
 - created_by (FK → users.user_id)
-

ratings

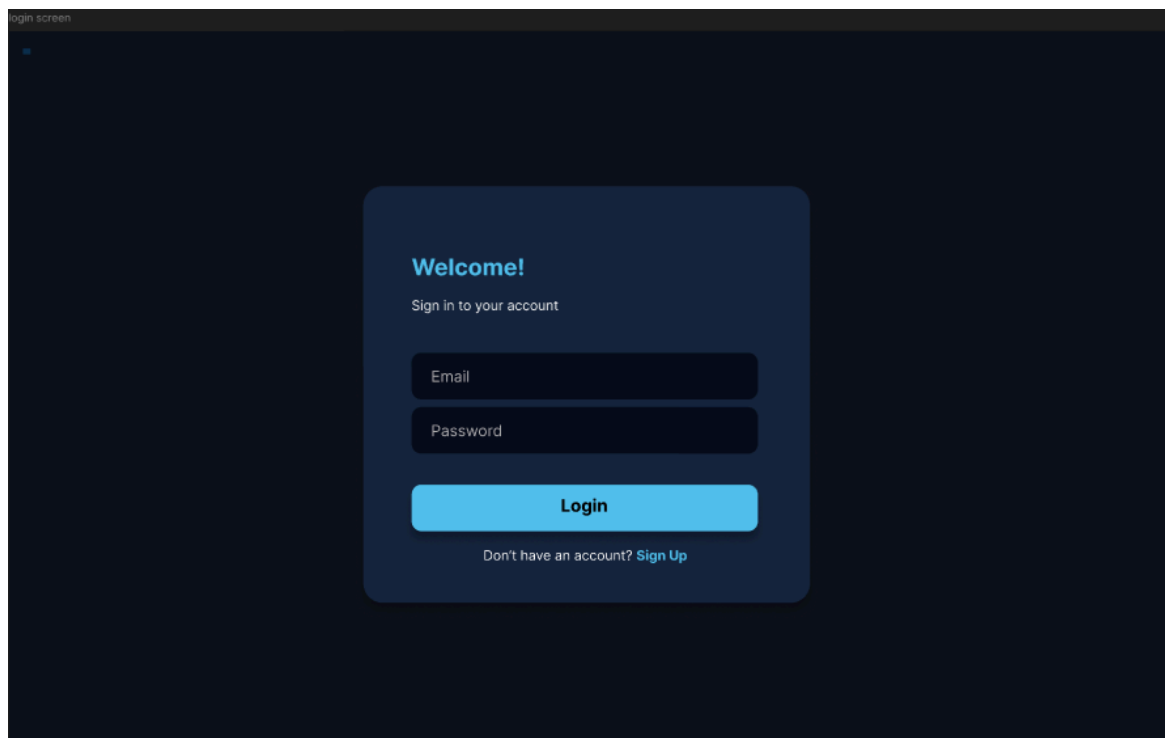
- ratingID (PK)
- serviceID (FK → services.service_id)

- userID(FK → users.user_id)
- starRate
- feedbackText
- dateCreated

7.0 UI/UX DESIGN

7.1 Web Application Wireframes

User Login Page



Header: Minimal / No Navigation Bar

Content: Authentication Card (Centered on screen)

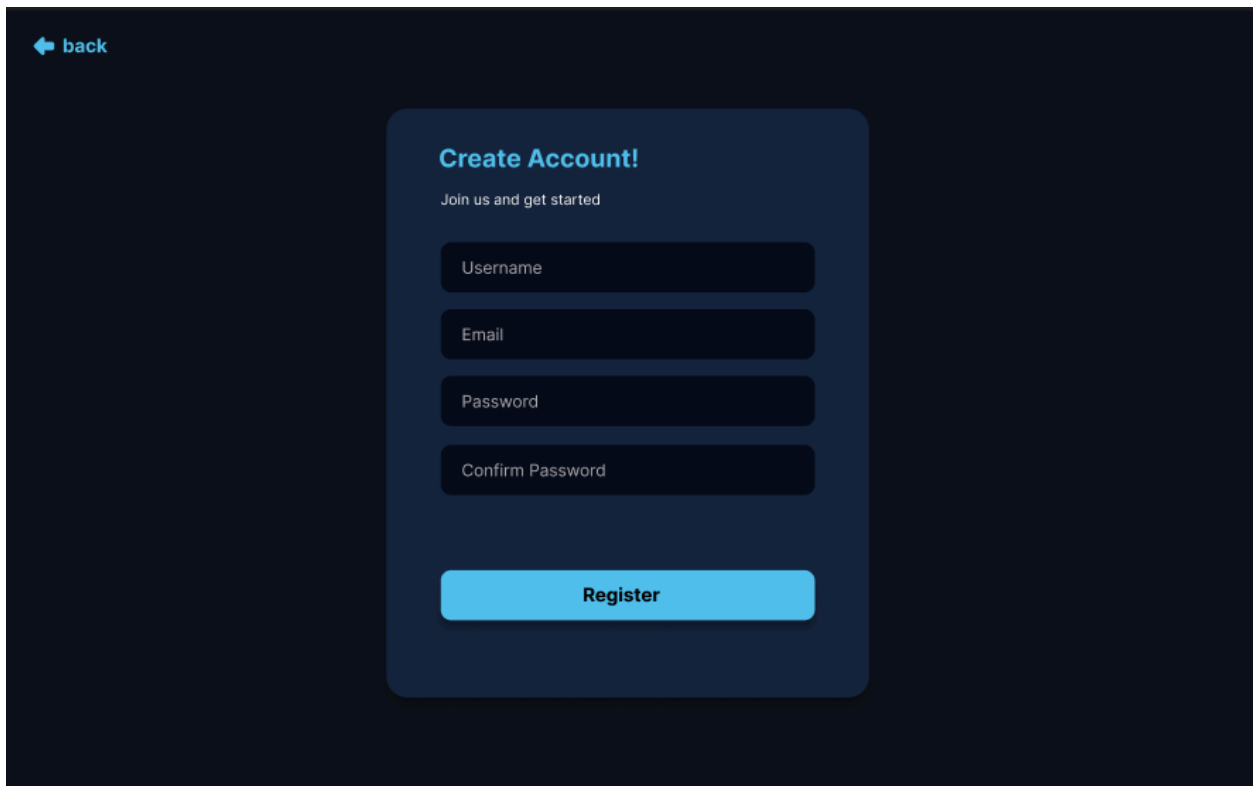
Authentication Card Components:

- Title: "Welcome!"
- Subtitle: "Sign in to your account"
- Email Input Field
- Password Input Field
- "Login" Button (Primary CTA)
- "Sign Up" Link (Secondary Action – below button)

Background: Dark gradient background (Full screen)

Footer: None (Clean authentication-focused layout)

User Registration Page

A dark-themed user registration page. In the top-left corner, there is a light blue back arrow icon followed by the text "back". Centered on the screen is a dark blue rounded rectangle card. Inside the card, the title "Create Account!" is displayed in light blue. Below it, the subtitle "Join us and get started" is in a smaller, lighter blue font. There are four input fields stacked vertically, each with a light blue placeholder text: "Username", "Email", "Password", and "Confirm Password". At the bottom of the card is a prominent light blue button with the text "Register" in dark blue.

Header: Back Button (Top-left corner)

Content (Centered Card Layout): Registration Card (Centered on screen)

Registration Card Components:

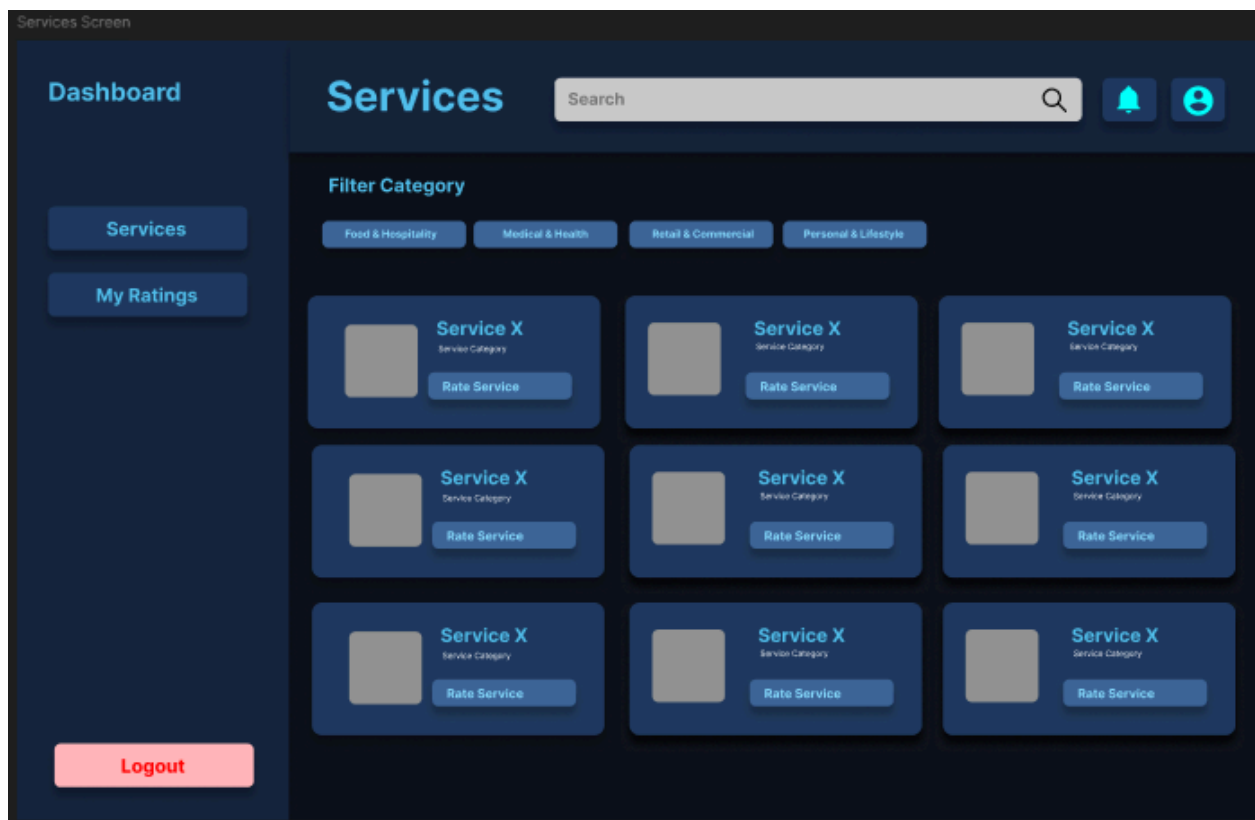
- Title: "Create Account!"
- Subtitle: "Join us and get started"
- Username Input Field
- Email Input Field

- Password Input Field
- Confirm Password Input Field
- "Register" Button (Primary CTA)

Background: Dark gradient background (Full screen)

Footer: None (Authentication-focused layout)

User Dashboard Page



Header:

Page Title ("Services") | Search Bar | Notification Icon | User Profile Icon

Sidebar:

Dashboard Title, Services, My Ratings, Logout Button

Content:

Filter Category Buttons (Food & Hospitality, Medical & Health, Retail & Commercial, Personal & Lifestyle)

Service Grid (3 columns desktop)

Each Service Card:

Image / Thumbnail

Service Name

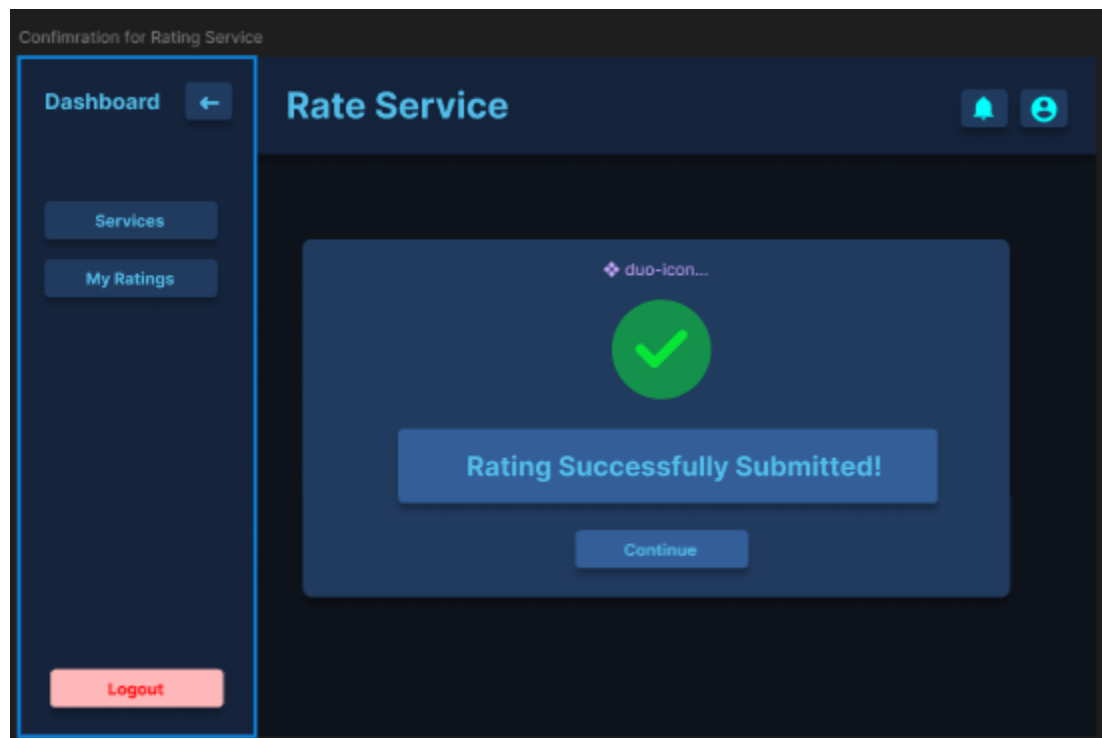
Service Category

"Rate Service" Button

Layout:

Left Sidebar + Top Navbar + Main Content Grid

Dark-themed UI

Rating Confirmation Page**Header:**

Page Title ("Rate Service") | Notification Icon | User Profile Icon

Sidebar:

Dashboard Title

- Services
- My Ratings
Logout Button

Content (Centered Card):

Success Icon (Checkmark)

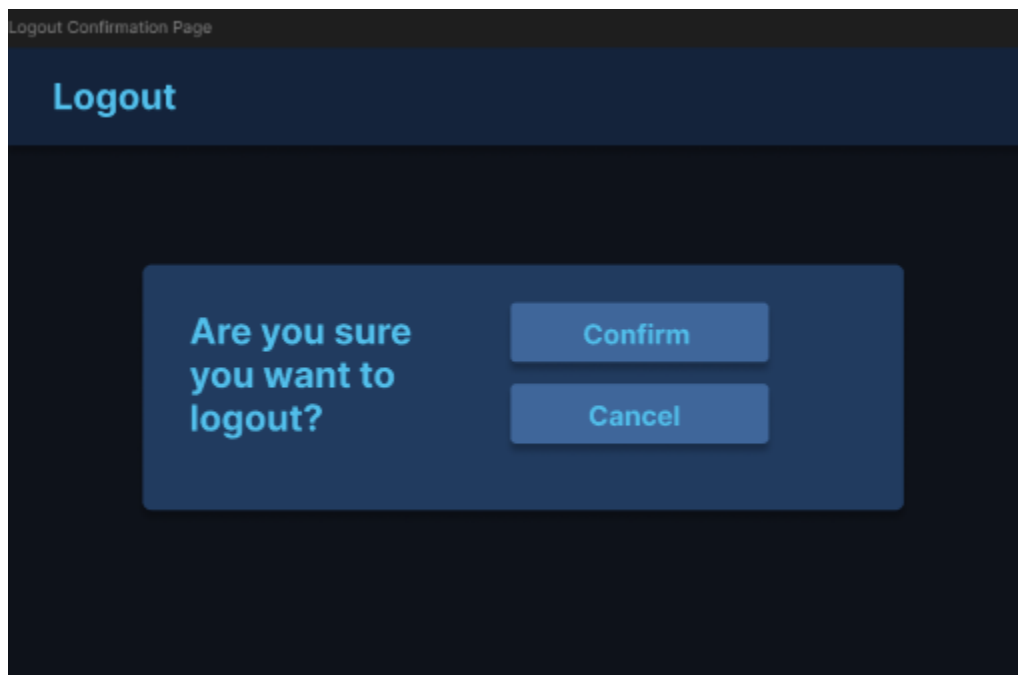
Message: "Rating Successfully Submitted!"

"Continue" Button

Layout:

Left Sidebar + Top Navbar + Centered Confirmation Card

Dark-themed UI

Logout Confirmation Page

Header:

Page Title ("Logout")

Content (Centered Card):

Message: "Are you sure you want to logout?"

Confirm Button

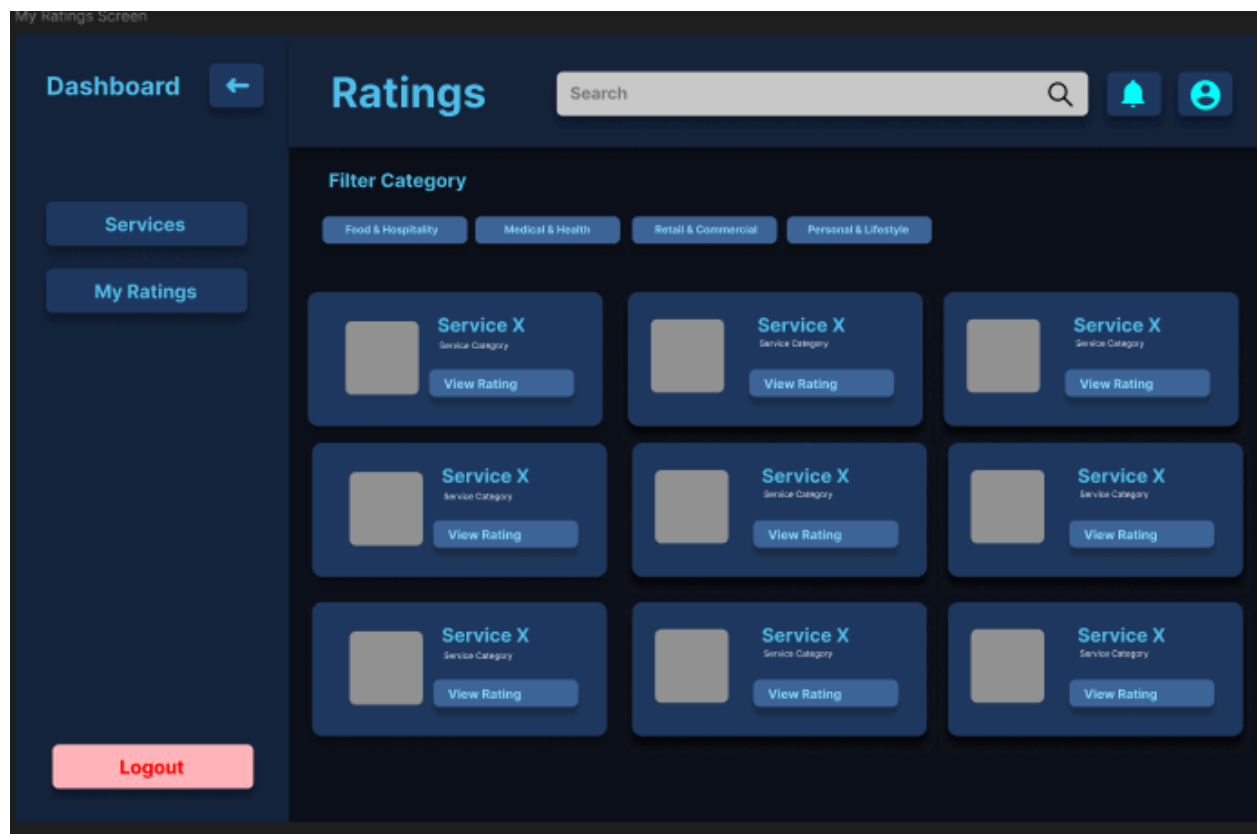
Cancel Button

Layout:

Top Header + Centered Confirmation Card

Dark-themed UI

My Ratings Page



Header:

Page Title ("Ratings") | Search Bar | Notification Icon | User Profile Icon

Sidebar:

Dashboard Title

- Services
- My Ratings (Active)
Logout Button

Content:

Filter Category Buttons (Food & Hospitality, Medical & Health, Retail & Commercial, Personal & Lifestyle)

Ratings Grid (3 columns desktop)

Each Rating Card:

Service Image / Thumbnail

Service Name

Service Category

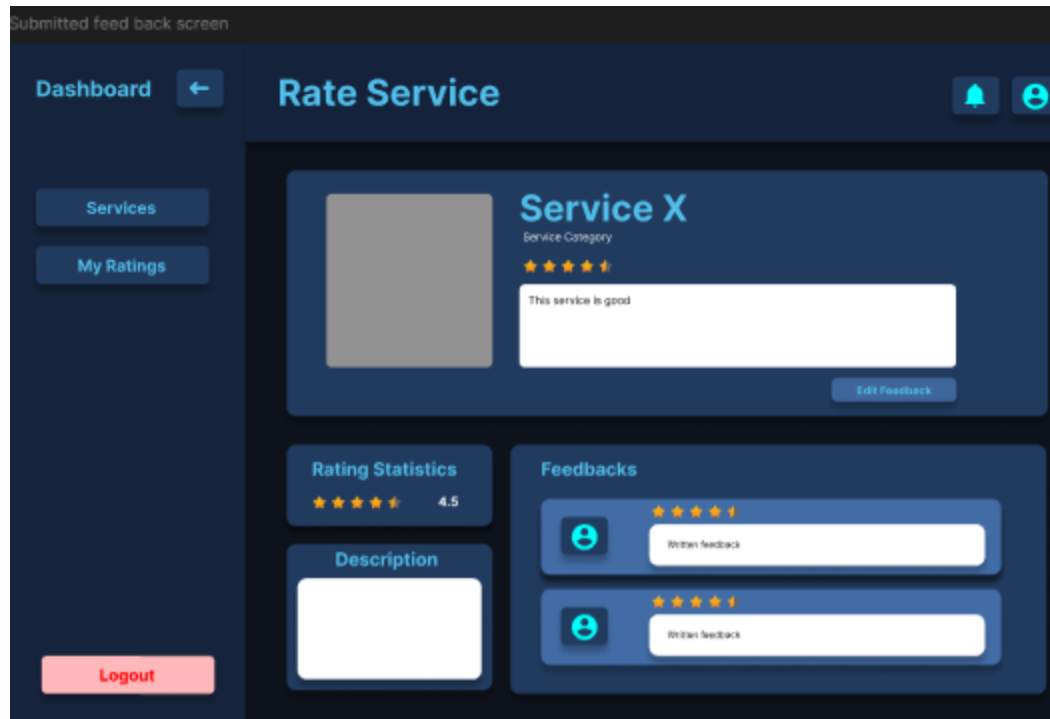
"View Rating" Button

Layout:

Left Sidebar + Top Navbar + Main Content Grid

Dark-themed UI

View Submitted Rating Page



Header:

Page Title ("Rate Service") | Notification Icon | User Profile Icon

Sidebar:

Dashboard Title

- Services
- My Ratings
- Logout Button

Content:

Service Details Section:

Service Image / Thumbnail

Service Name

Service Category

User Feedback Text

"Edit Feedback" Button

Rating Statistics Section:

Star Rating Display

Average Rating Value

Description Section:

Service Description Box

Feedbacks Section:

List of User Feedback Cards

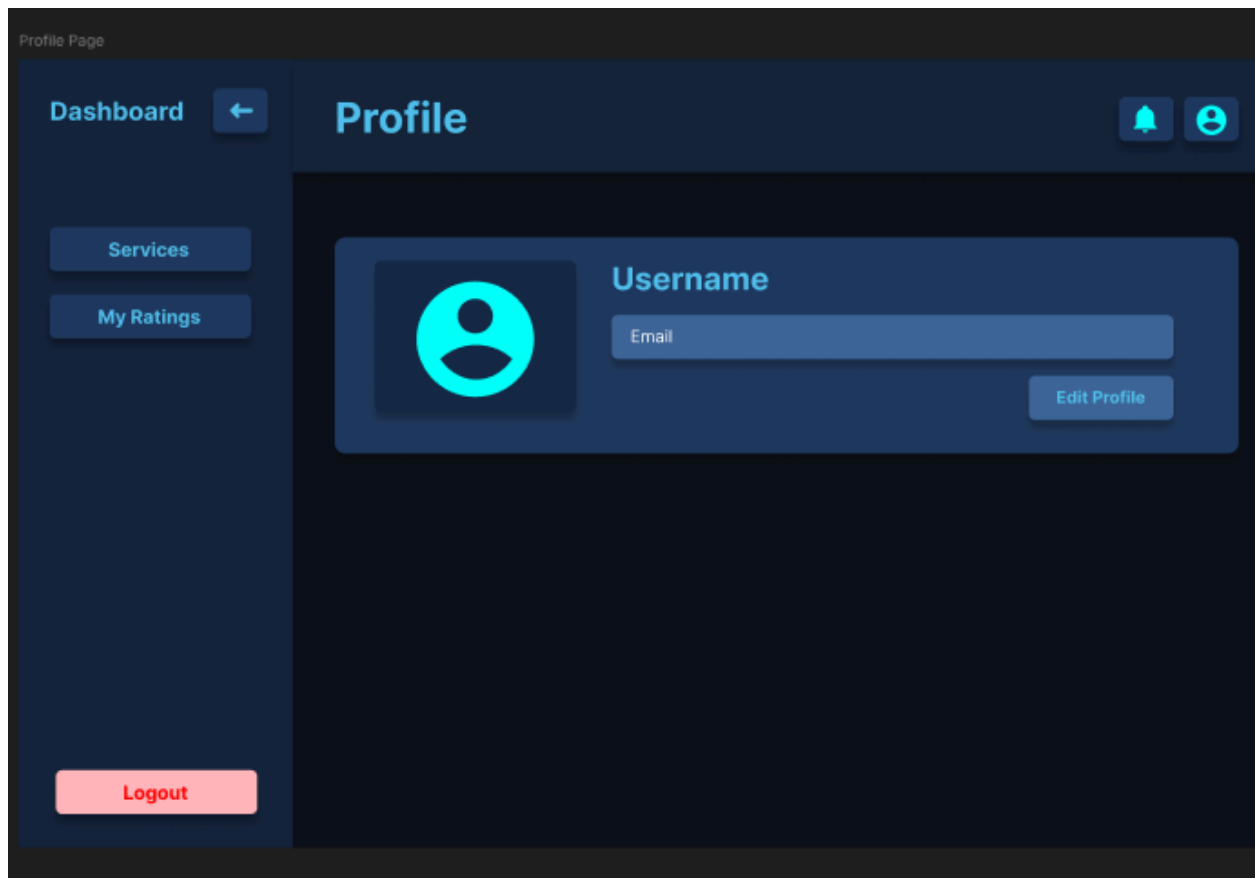
- User Icon
- Star Rating
- Written Feedback

Layout:

Left Sidebar + Top Navbar + Multi-section Content Cards

Dark-themed UI

User/admin Profile Page



Header:

Page Title ("Profile") | Notification Icon | User Profile Icon

Sidebar:

Dashboard Title

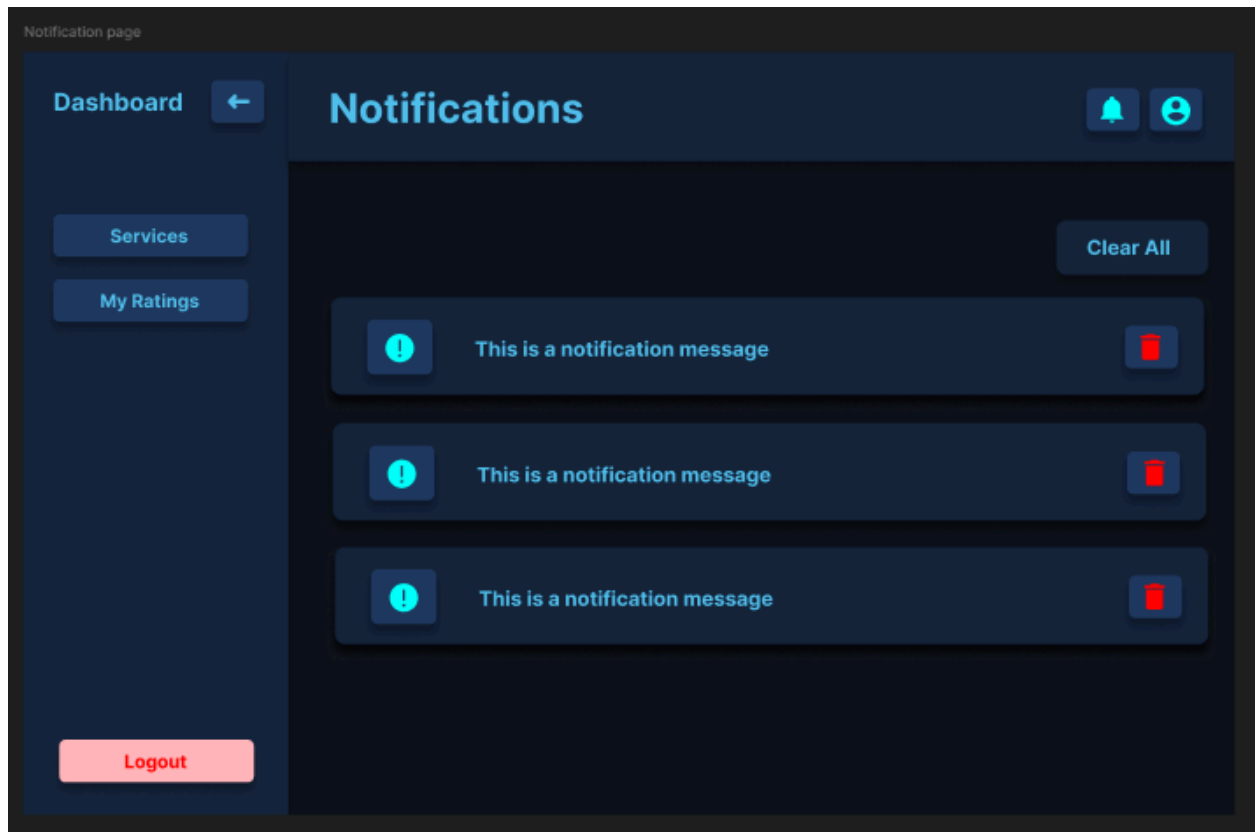
- Services
- My Ratings
- Logout Button

Content (Profile Card):

User Avatar / Profile Icon
Username Display
Email Display Field
"Edit Profile" Button

Layout:

Left Sidebar + Top Navbar + Centered Profile Card
Dark-themed UI

Notification Page**Header:**

Page Title ("Notifications") | Notification Icon | User Profile Icon

Sidebar:

Dashboard Title

- Services
- My Ratings
- Logout Button

Content:

"Clear All" Button

Notification List (Vertical Layout)

Each Notification Card:

Notification Icon

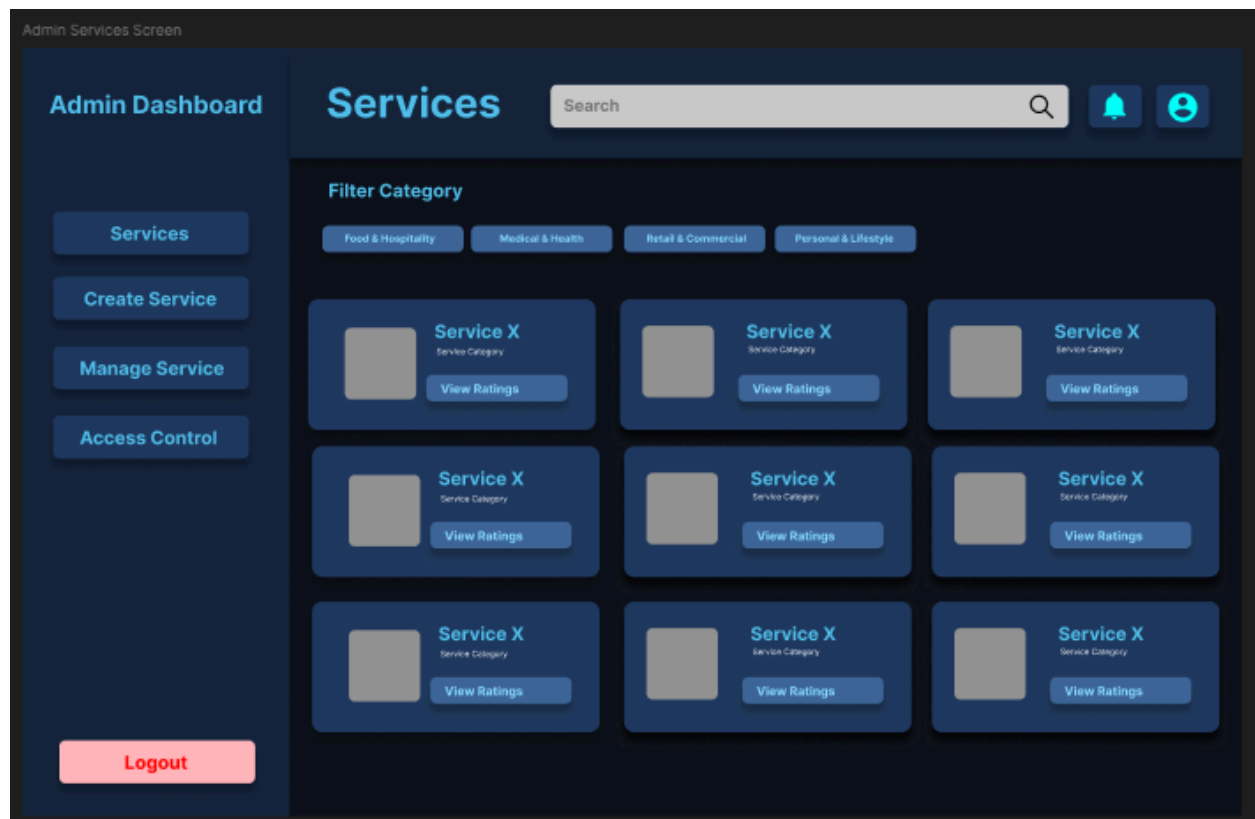
Notification Message

Delete Icon

Layout:

Left Sidebar + Top Navbar + Stacked Notification Cards

Dark-themed UI

Admin Services Page**Header:**

Page Title ("Services") | Search Bar | Notification Icon | User Profile Icon

Sidebar:

Admin Dashboard Title

- Services (Active)
- Create Service
- Manage Service
- Access Control
- Logout Button

Content:

Filter Category Buttons (Food & Hospitality, Medical & Health, Retail & Commercial, Personal & Lifestyle)

Services Grid (3 columns desktop)

Each Service Card:

Service Image / Thumbnail

Service Name

Service Category

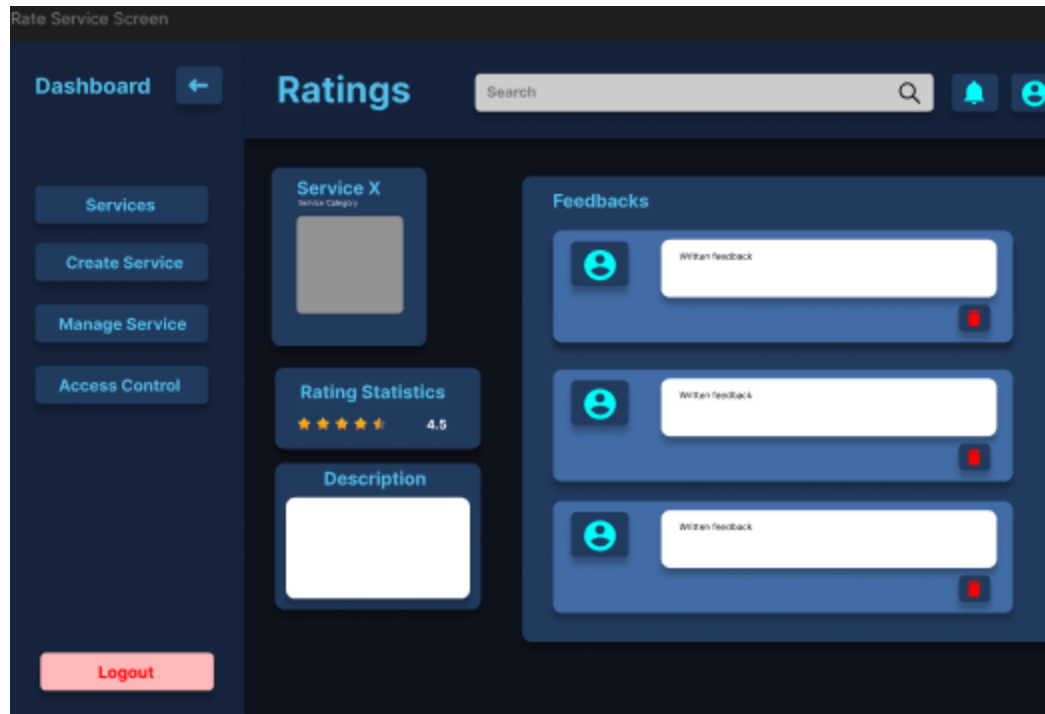
"View Ratings" Button

Layout:

Left Admin Sidebar + Top Navbar + Main Content Grid

Dark-themed UI

Admin View Feedbacks Page



Header:

Page Title ("Ratings") | Search Bar | Notification Icon | User Profile Icon

Sidebar:

Admin Dashboard Title

- Services
- Create Service
- Manage Service
- Access Control
- Logout Button

Content:

Service Overview Section:

Service Image / Thumbnail

Service Name

Service Category

Rating Statistics Section:

Star Rating Display

Average Rating Value

Description Section:

Service Description Box

Feedbacks Section (Vertical List):

Feedback Cards

Each Feedback Card:

User Icon

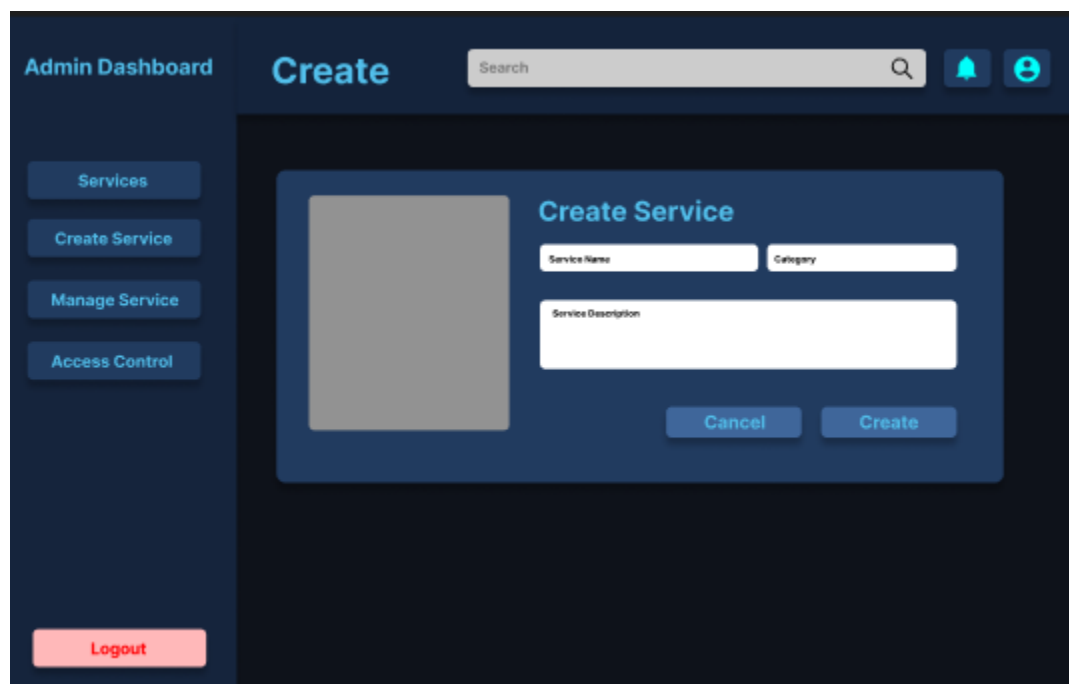
Written Feedback

Delete Button

Layout:

Left Admin Sidebar + Top Navbar + Multi-section Content Layout

Dark-themed UI

Admin Create Service Page

The screenshot displays the 'Admin Create Service Page' within a dark-themed dashboard. The top header includes 'Admin Dashboard' and 'Create' in light blue, followed by a search bar and notification/user icons. A left sidebar contains buttons for 'Services', 'Create Service', 'Manage Service', 'Access Control', and a red 'Logout' button at the bottom. The main content area features a 'Create Service' modal with a placeholder image, input fields for 'Service Name' and 'Category', a larger 'Service Description' field, and 'Cancel' and 'Create' buttons.

Header:

Page Title ("Create") | Search Bar | Notification Icon | User Profile Icon

Sidebar:

Admin Dashboard Title

- Services
- Create Service (Active)
- Manage Service
- Access Control
- Logout Button

Content (Centered Form Card):

Service Image Upload / Placeholder

Form Fields:

Service Name Input

Category Dropdown

Service Description Text Area

Actions:

Cancel Button

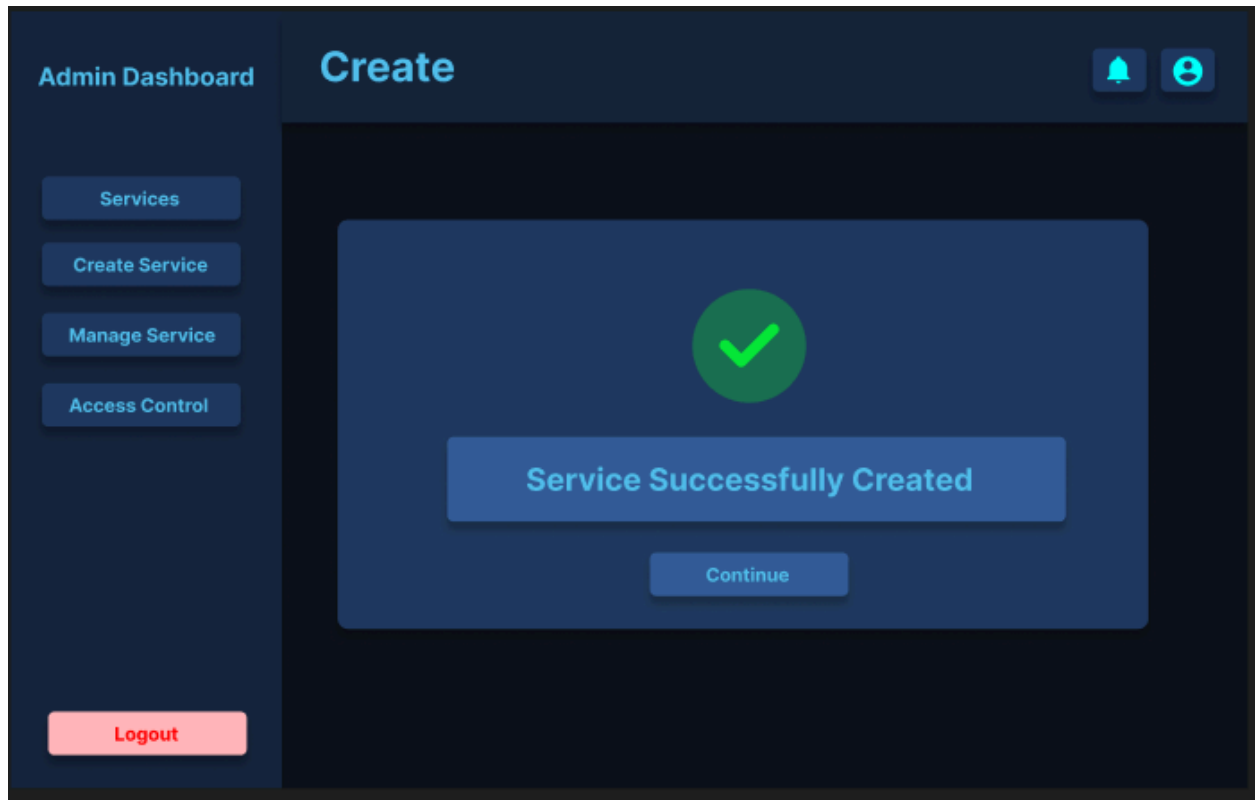
Create Button

Layout:

Left Admin Sidebar + Top Navbar + Centered Form Card

Dark-themed UI

Admin Service Creation Confirmation Page

**Header:**

Page Title ("Create") | Notification Icon | User Profile Icon

Sidebar:

Admin Dashboard Title

- Services
- Create Service
- Manage Service
- Access Control
- Logout Button

Content (Centered Confirmation Card):

Success Icon (Checkmark)

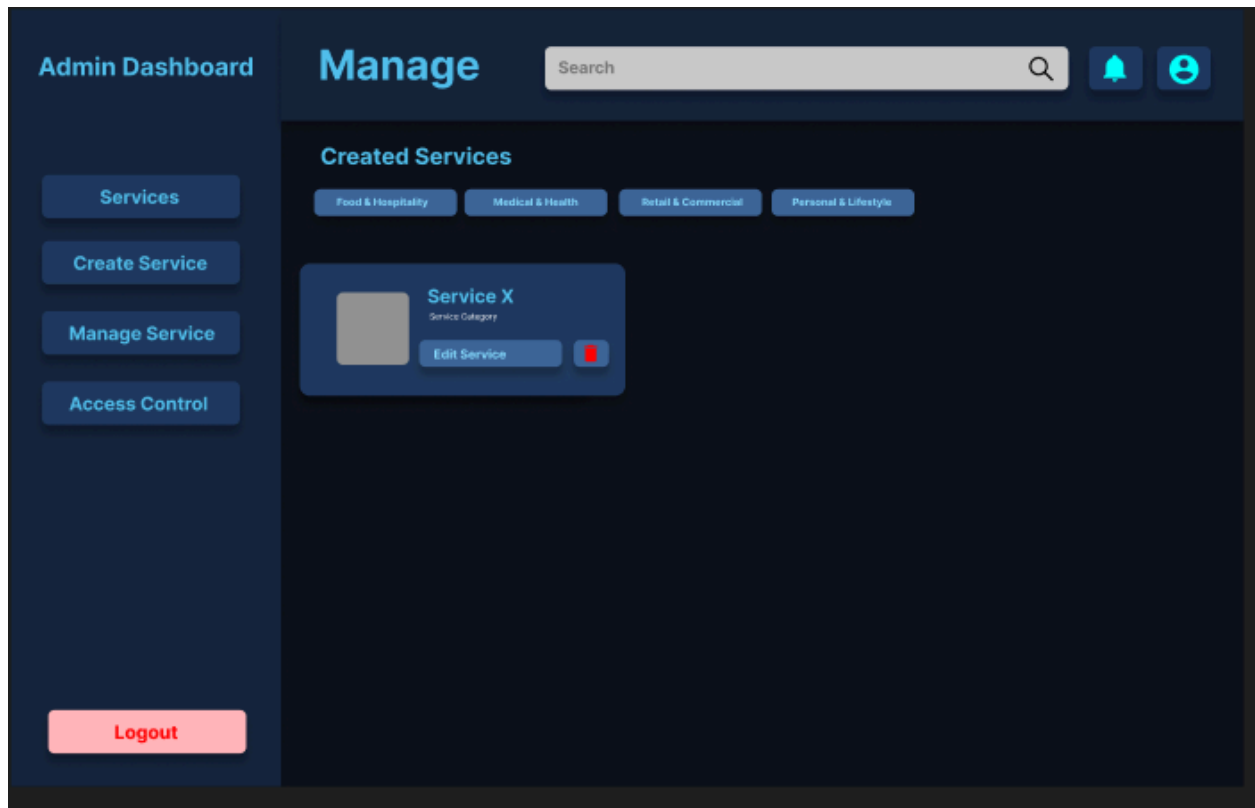
Message: "Service Successfully Created"

"Continue" Button

Layout:

Left Admin Sidebar + Top Navbar + Centered Confirmation Card
Dark-themed UI

Admin Manage Service Page



Header:

Page Title ("Manage") | Search Bar | Notification Icon | User Profile Icon

Sidebar:

Admin Dashboard Title

- Services
- Create Service
- Manage Service (Active)

- Access Control
Logout Button

Content:

Section Title: "Created Services"

Filter Category Buttons (Food & Hospitality, Medical & Health, Retail & Commercial, Personal & Lifestyle)

Services List / Grid

Each Service Card:

Service Image / Thumbnail

Service Name

Service Category

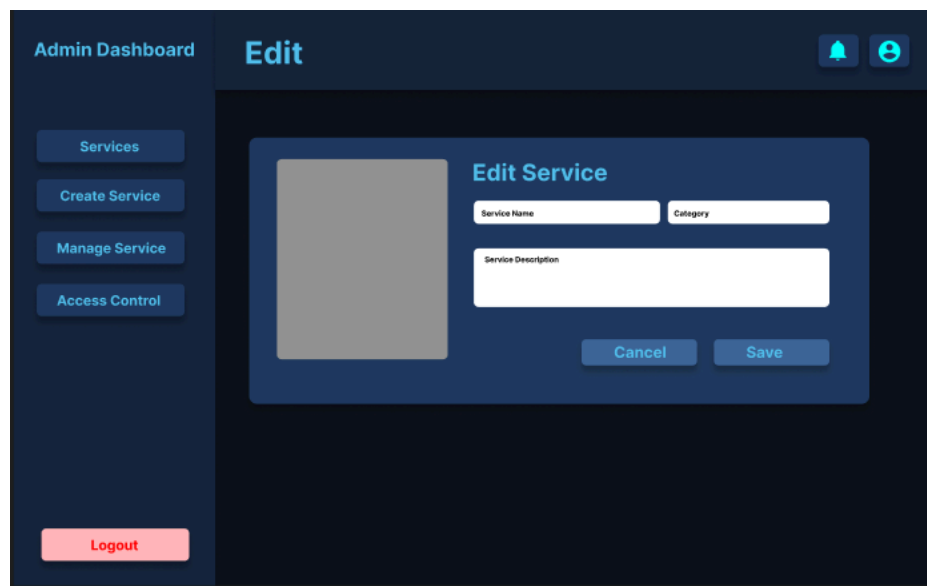
"Edit Service" Button

Delete Icon

Layout:

Left Admin Sidebar + Top Navbar + Service Management Grid

Dark-themed UI

Admin Edit Service Page

The screenshot displays the 'Admin Edit Service Page' within a dark-themed dashboard. On the left, a sidebar contains the 'Admin Dashboard' header and a list of navigation buttons: 'Services', 'Create Service', 'Manage Service', 'Access Control', and a red 'Logout' button at the bottom. The main content area is titled 'Edit' and features a notification bell and user profile icon in the top right. The central 'Edit Service' form includes a placeholder for a service image, input fields for 'Service Name' and 'Category', a larger text area for 'Service Description', and 'Cancel' and 'Save' buttons at the bottom.

Header:

Page Title ("Edit") | Notification Icon | User Profile Icon

Sidebar:

Admin Dashboard Title

- Services
- Create Service
- Manage Service
- Access Control
- Logout Button

Content (Centered Form Card):

Service Image Display / Upload Option

Form Fields:

Service Name Input (Pre-filled)

Category Dropdown (Pre-selected)

Service Description Text Area (Pre-filled)

Actions:

Cancel Button

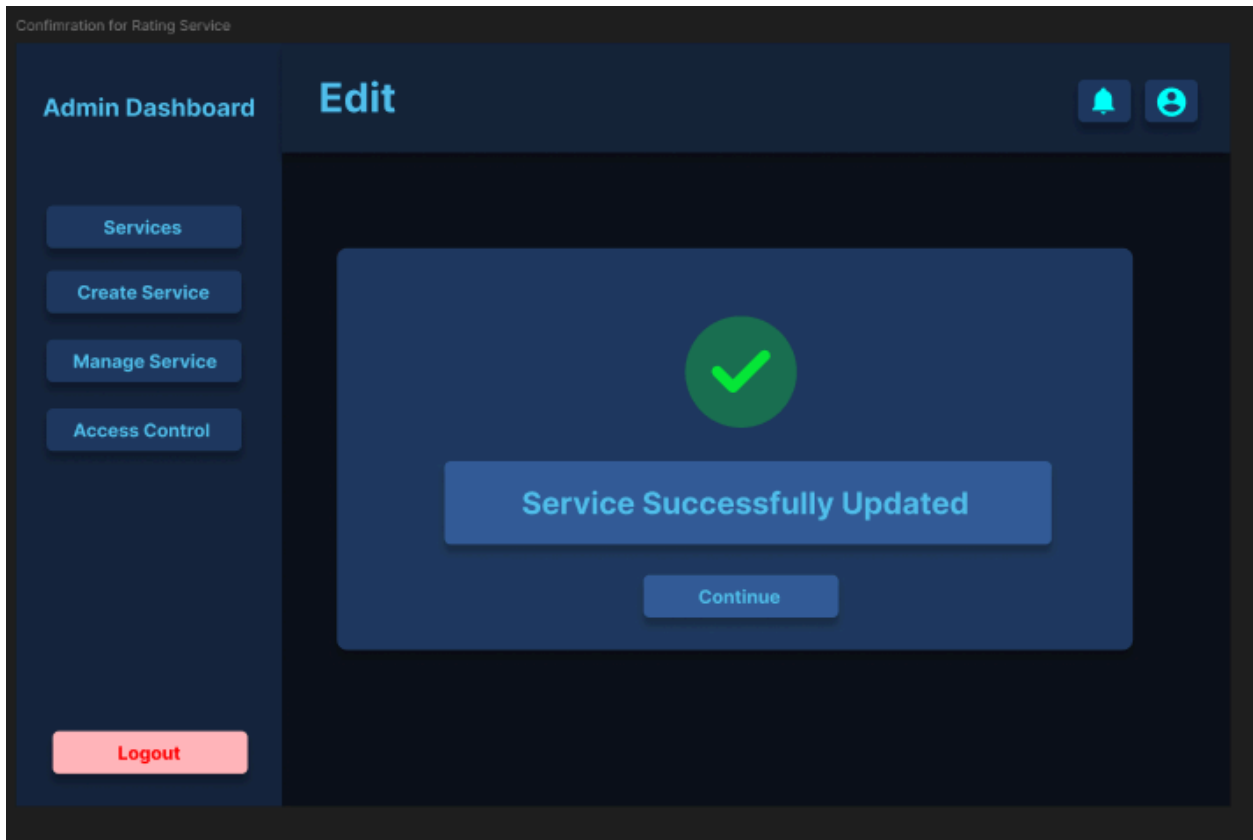
Save Button

Layout:

Left Admin Sidebar + Top Navbar + Centered Edit Form Card

Dark-themed UI

Admin Edit Service Confirmation Page



Header:

Page Title ("Edit") | Notification Icon | User Profile Icon

Sidebar:

Admin Dashboard Title

- Services
 - Create Service
 - Manage Service
 - Access Control
- Logout Button

Content (Centered Confirmation Card):

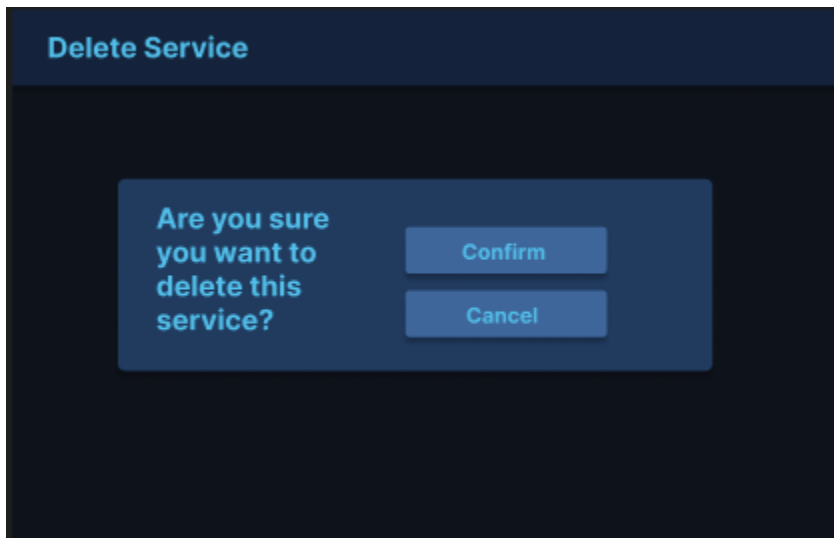
Success Icon (Checkmark)

Message: "Service Successfully Updated"

"Continue" Button

Layout:

Left Admin Sidebar + Top Navbar + Centered Confirmation Card
Dark-themed UI

Admin Delete Service Confirmation Page**Header:**

Page Title ("Delete Service")

Content (Centered Confirmation Card):

Message: "Are you sure you want to delete this service?"

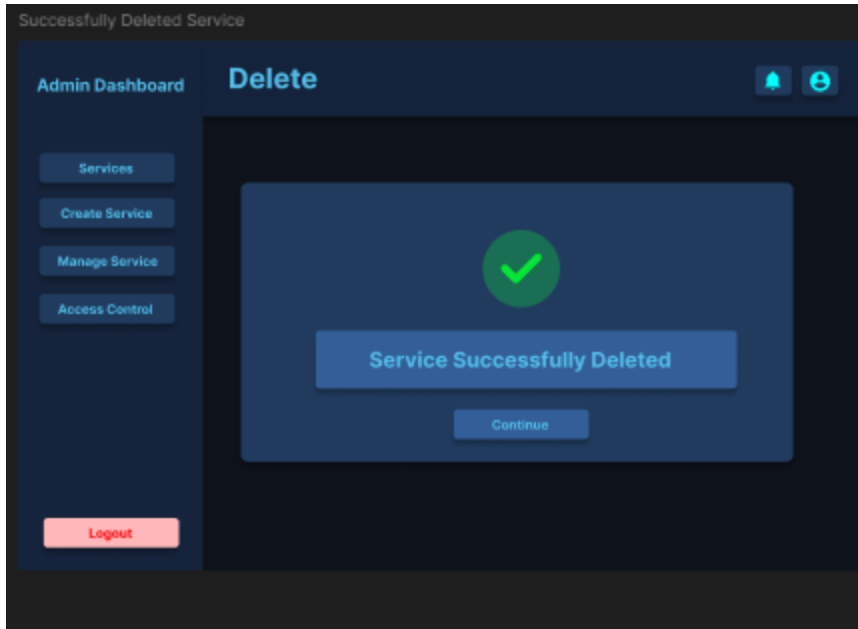
Confirm Button

Cancel Button

Layout:

Top Header + Centered Confirmation Card
Dark-themed UI

Admin Service Deletion Success Page

**Header:**

Page Title ("Delete") | Notification Icon | User Profile Icon

Sidebar:

Admin Dashboard Title

- Services
 - Create Service
 - Manage Service
 - Access Control
- Logout Button

Content (Centered Confirmation Card):

Success Icon (Checkmark)

Message: "Service Successfully Deleted"

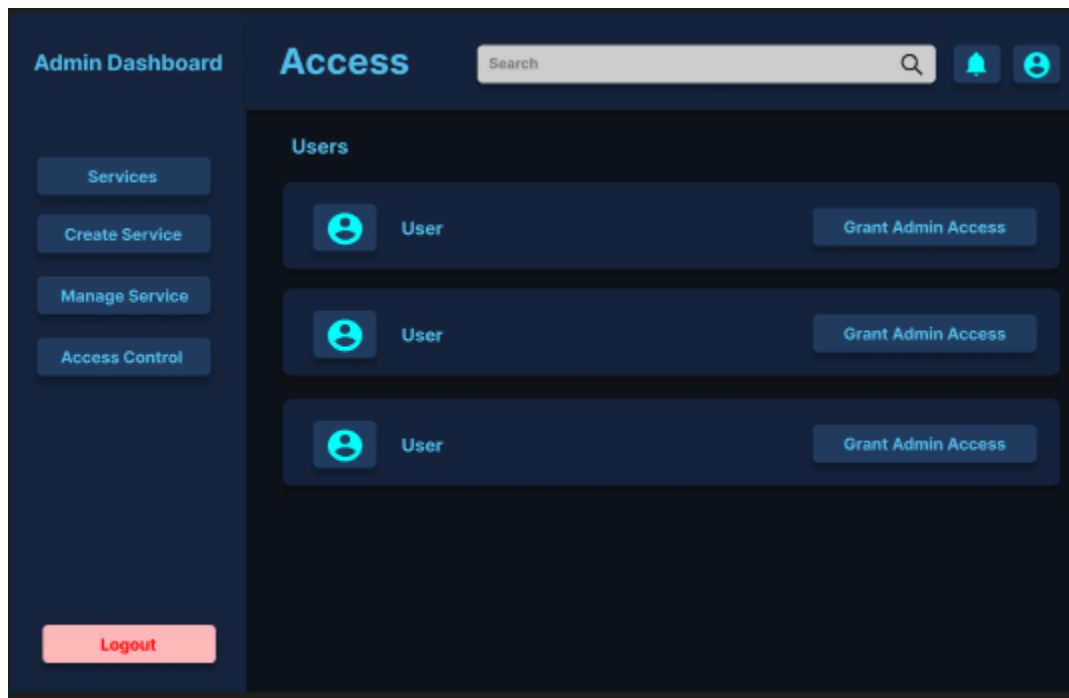
"Continue" Button

Layout:

Left Admin Sidebar + Top Navbar + Centered Confirmation Card

Dark-themed UI

Admin Access Control Page



Header:

Page Title ("Access") | Search Bar | Notification Icon | User Profile Icon

Sidebar:

Admin Dashboard Title

- Services
- Create Service
- Manage Service
- Access Control (Active)
- Logout Button

Content:

Section Title: "Users"

Users List (Vertical Layout)

Each User Card:

User Icon
Username
"Grant Admin Access" Button

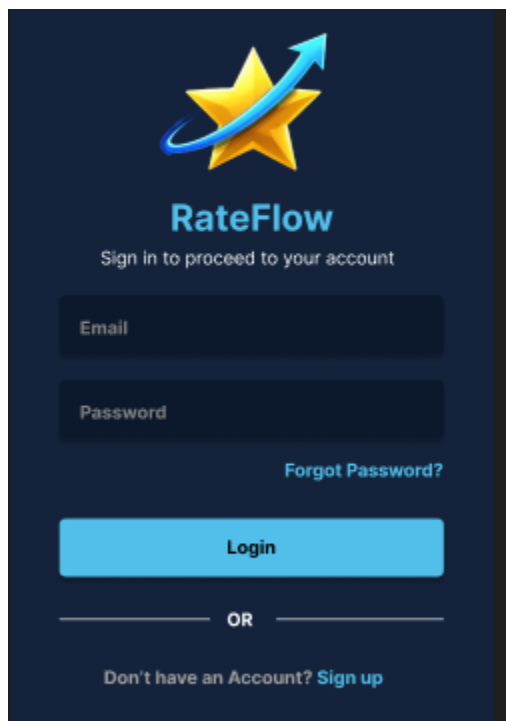
Layout:

Left Admin Sidebar + Top Navbar + Stacked User Cards
Dark-themed UI

7.2 Mobile Application Wireframes

Note: This should be wireframes from Figma

Mobile Login Page



Header:

App Logo
App Name ("RateFlow")
Subtitle: "Sign in to proceed to your account"

Content (Single Column Layout):

Email Input Field

Password Input Field
"Forgot Password?" Link

Primary Button:
"Login" Button

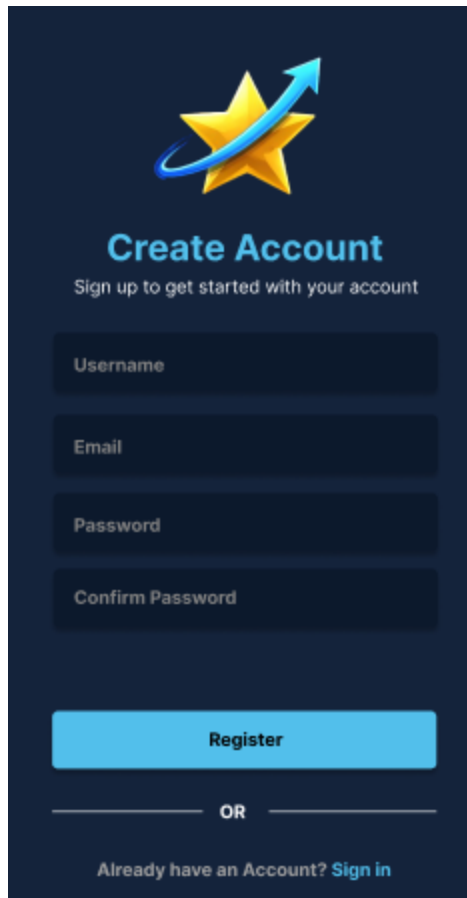
Divider:
"OR" Separator Line

Secondary Action:
"Don't have an Account? Sign up" Link

Layout:

Centered Vertical Layout (Single Column)
Full-screen Dark-themed UI
Mobile-optimized spacing and touch-friendly buttons

Mobile Registration Page

A dark-themed 'Create Account' form. At the top is a logo featuring a yellow star with a blue arrow curving around it. Below the logo is the title 'Create Account' in a light blue font, followed by the subtitle 'Sign up to get started with your account' in a smaller, lighter font. The form contains four input fields: 'Username', 'Email', 'Password', and 'Confirm Password', each with a light blue placeholder text. Below these fields is a prominent light blue 'Register' button. Underneath the button is a horizontal line with the word 'OR' in the center. At the bottom, there is a link that says 'Already have an Account? Sign in'.

Header:

App Logo

Title: "Create Account"

Subtitle: "Sign up to get started with your account"

Content (Single Column Layout):

Username Input Field

Email Input Field

Password Input Field

Confirm Password Input Field

Primary Button:

"Register" Button

Divider:

"OR" Separator Line

Secondary Action:

"Already have an Account? Sign in" Link

Layout:

Centered Vertical Layout (Single Column)

Full-screen Dark-themed UI

Mobile-optimized spacing and touch-friendly buttons

Mobile User Dashboard Page



Header:

Menu Icon | App Name ("RateFlow") | Notification Icon | User Profile Icon

Content (Scrollable Single Column):

Welcome Card:

App Logo

Message: "Welcome user!"

Filter Section:

Filter Button

Search Bar

Services Grid (2 Columns Mobile Layout):**Each Service Card:**

Service Image / Thumbnail

Service Name

Service Category

"Rate Service" Button

Quick Actions Section:

"View All" Button

Layout:

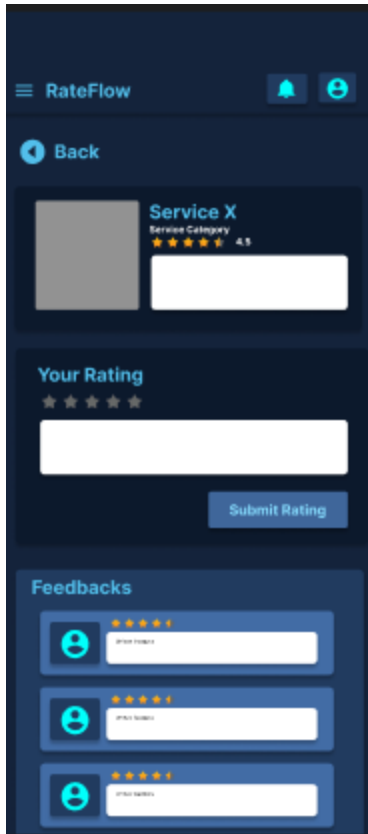
Top Navbar + Scrollable Content

2-Column Service Grid

Dark-themed UI

Mobile-optimized spacing

Mobile Submit Rating Page



Header:

Menu Icon | App Name ("RateFlow") | Notification Icon | User Profile Icon

Back Button (Below Header)

Content (Scrollable Single Column):

Service Details Section:

Service Image / Thumbnail

Service Name

Service Category

Average Rating Display (Stars + Value)

Your Rating Section:

Star Rating Input

Feedback Text Area

"Submit Rating" Button

Feedbacks Section:

Feedback List (Vertical Layout)

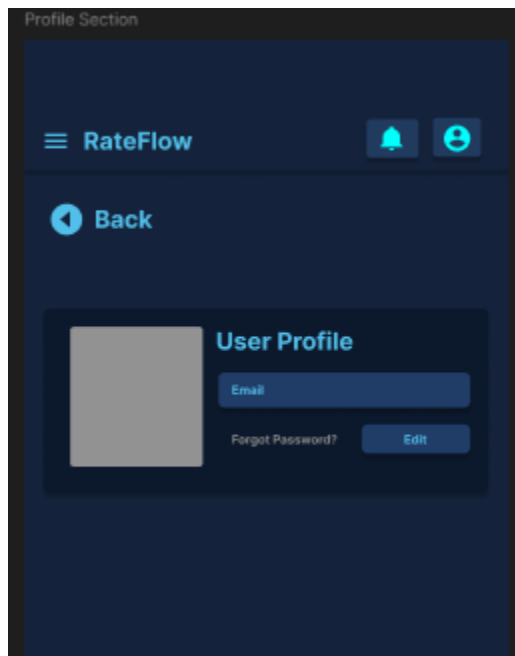
Each Feedback Card:

User Icon
Star Rating
Written Feedback

Layout:

Top Navbar + Scrollable Content
Single Column Layout
Dark-themed UI
Mobile-optimized spacing and touch-friendly controls

Mobile User Profile Page



Header:

Menu Icon | App Name ("RateFlow") | Notification Icon | User Profile Icon

Back Button (Below Header)

Content (Single Column Layout):

Profile Card:

User Avatar / Profile Image

Title: "User Profile"
Email Display Field

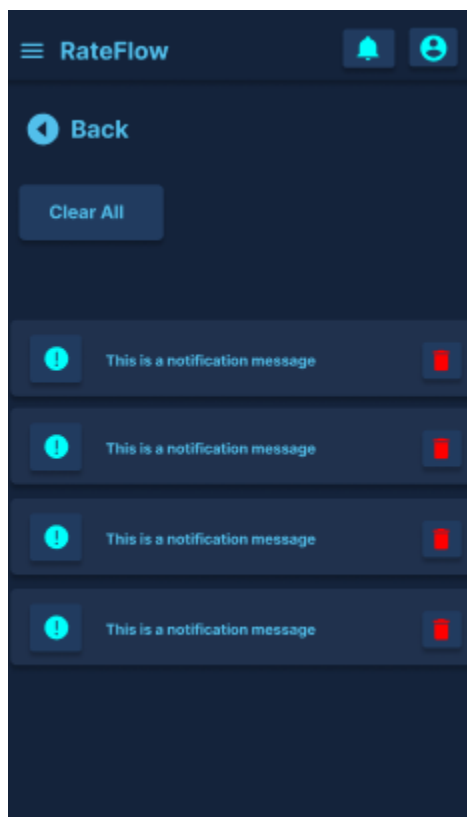
Actions:

"Forgot Password?" Link
"Edit" Button

Layout:

Top Navbar + Scrollable Content
Centered Profile Card
Dark-themed UI
Mobile-optimized spacing

Mobile Notifications Page



Header:

Menu Icon | App Name ("RateFlow") | Notification Icon | User Profile Icon

Back Button (Below Header)

Content (Scrollable Single Column):

"Clear All" Button

Notifications List (Vertical Layout)

Each Notification Card:

Notification Icon

Notification Message

Delete Icon

Layout:

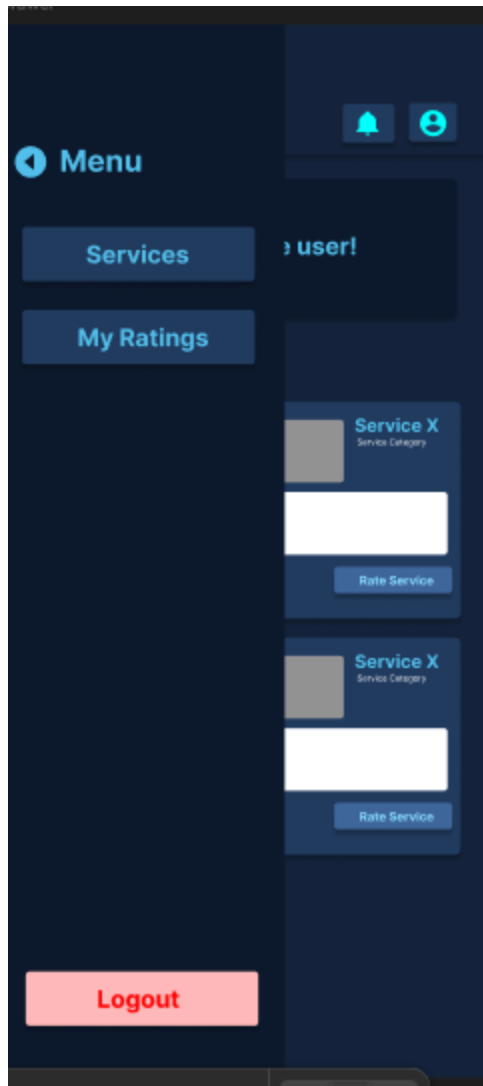
Top Navbar + Scrollable Content

Stacked Notification Cards

Dark-themed UI

Mobile-optimized spacing and touch-friendly controls

Mobile User Drawer (Side Menu)



Header:

Title: "Menu"

Menu Items:

Services

My Ratings

Action:

Logout Button (Bottom)

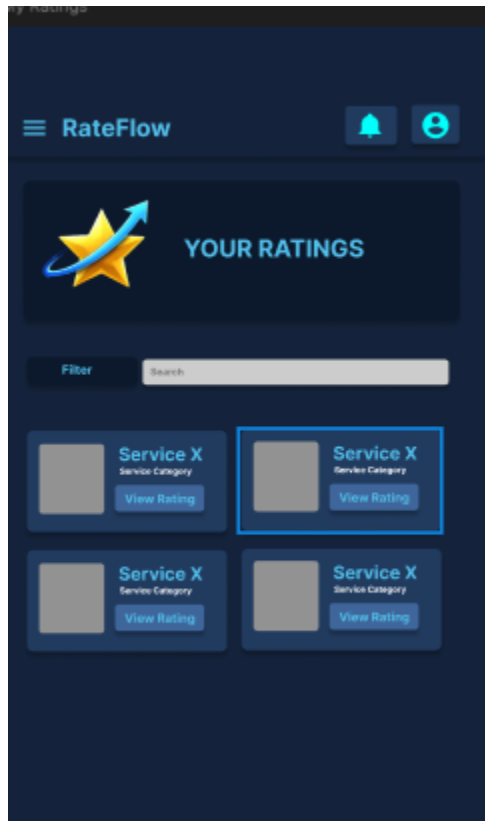
Layout:

Slide-in Left Drawer

Overlay on Main Content

Dark-themed UI
Touch-friendly Navigation Buttons

Mobile My Ratings Page



Header:

Menu Icon | App Name ("RateFlow") | Notification Icon | User Profile Icon

Content (Scrollable Single Column):

Banner Section:

App Logo

Title: "Your Ratings"

Filter Section:

Filter Button

Search Bar

Ratings Grid (2 Columns Mobile Layout):

Each Rating Card:

Service Image / Thumbnail

Service Name

Service Category

"View Rating" Button

Layout:

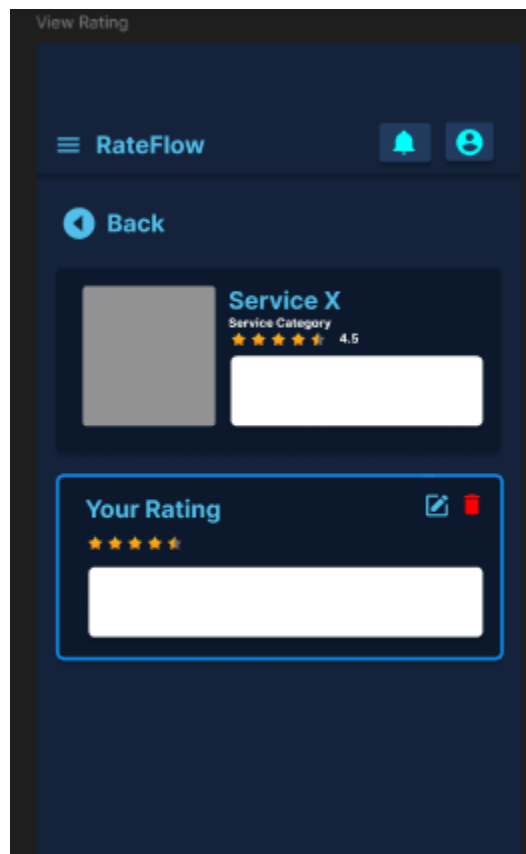
Top Navbar + Scrollable Content

2-Column Card Grid

Dark-themed UI

Mobile-optimized spacing and touch-friendly controls

Mobile View Rating Page



Header:

Menu Icon | App Name ("RateFlow") | Notification Icon | User Profile Icon

Back Button (Below Header)

Content (Scrollable Single Column):

Service Details Section:

- Service Image / Thumbnail
- Service Name
- Service Category
- Average Rating Display (Stars + Value)
- Service Description Box

Your Rating Section:

- Star Rating Display
- Written Feedback

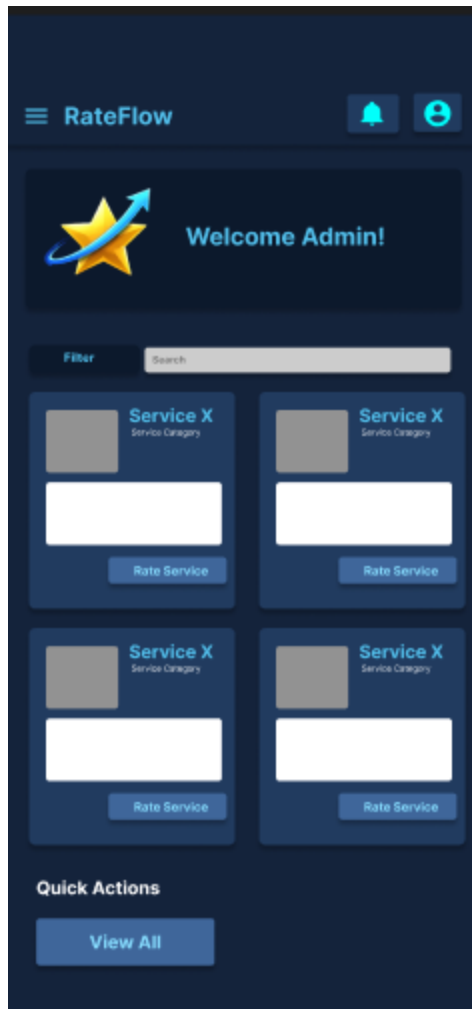
Actions:

- Edit Icon
- Delete Icon

Layout:

- Top Navbar + Scrollable Content
- Single Column Layout
- Dark-themed UI
- Mobile-optimized spacing and touch-friendly controls

Mobile Admin Dashboard Page



Header:

Menu Icon | App Name ("RateFlow") | Notification Icon | User Profile Icon

Content (Scrollable Single Column):

Welcome Banner:

App Logo

Message: "Welcome Admin!"

Filter Section:

Filter Button

Search Bar

Services Grid (2 Columns Mobile Layout):

Each Service Card:

Service Image / Thumbnail

Service Name
Service Category
"Rate Service" Button

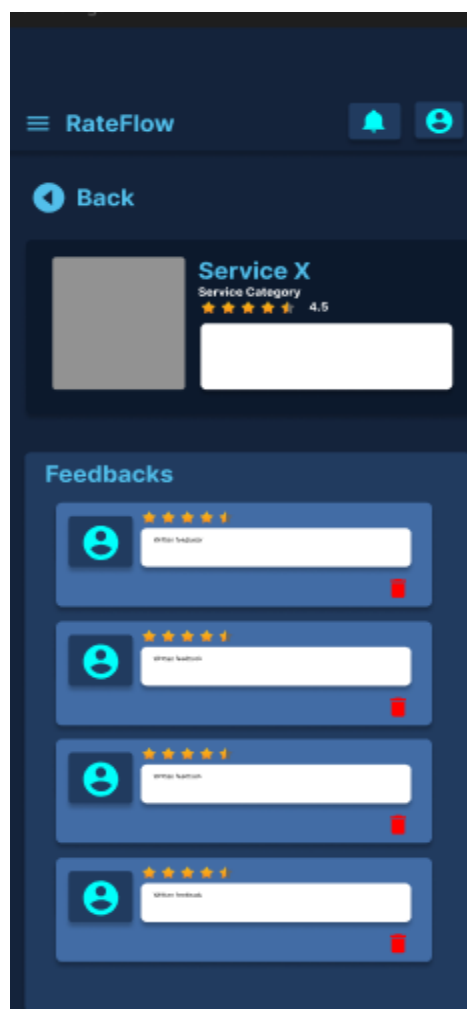
Quick Actions Section:

"View All" Button

Layout:

Top Navbar + Scrollable Content
2-Column Card Grid
Dark-themed UI
Mobile-optimized spacing and touch-friendly controls

Mobile Admin – Manage Submitted Ratings Page



Header:

Menu Icon | App Name ("RateFlow") | Notification Icon | User Profile Icon

Back Button (Below Header)

Content (Scrollable Single Column):**Service Details Section:**

Service Image / Thumbnail

Service Name

Service Category

Average Rating Display (Stars + Value)

Service Description Box

Feedbacks Section:

Feedback List (Vertical Layout)

Each Feedback Card:

User Icon

Star Rating

Written Feedback

Delete Icon

Layout:

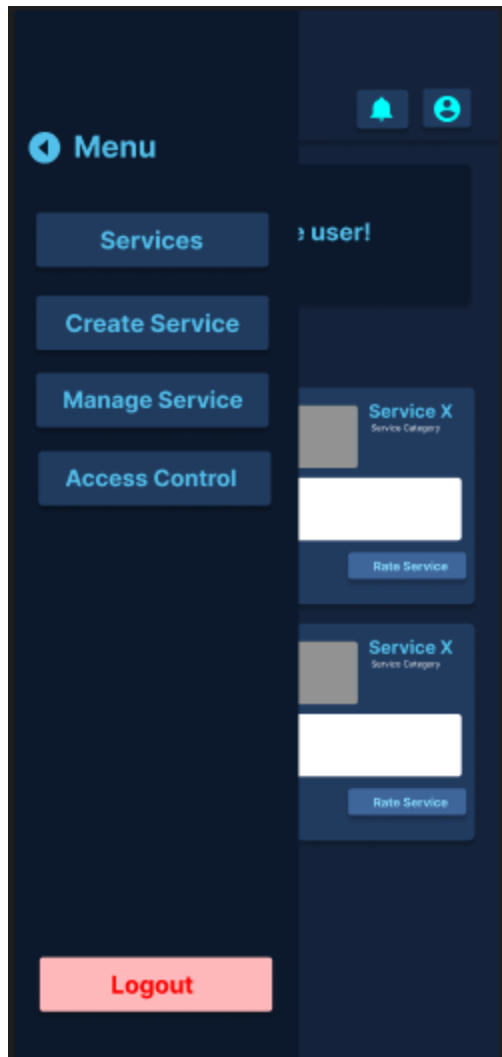
Top Navbar + Scrollable Content

Single Column Layout

Dark-themed UI

Mobile-optimized spacing and touch-friendly controls

Mobile Admin Drawer (Side Menu)



Header:

Title: "Menu"

Menu Items:

Services
Create Service
Manage Service
Access Control

Action:

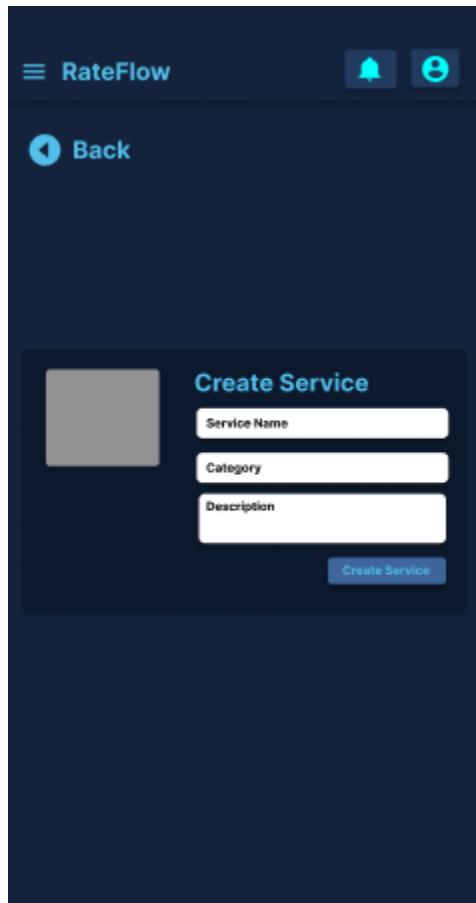
Logout Button (Bottom)

Layout:

Slide-in Left Drawer
Overlay on Main Content

Dark-themed UI
Touch-friendly Navigation Buttons

Mobile Admin – Create Service Page



Header:

Menu Icon | App Name ("RateFlow") | Notification Icon | User Profile Icon

Back Button (Below Header)

Content (Single Column Layout):

Create Service Card:

Service Image Upload / Placeholder

Form Fields:

Service Name Input

Category Dropdown
Description Text Area

Action:

"Create Service" Button

Layout:

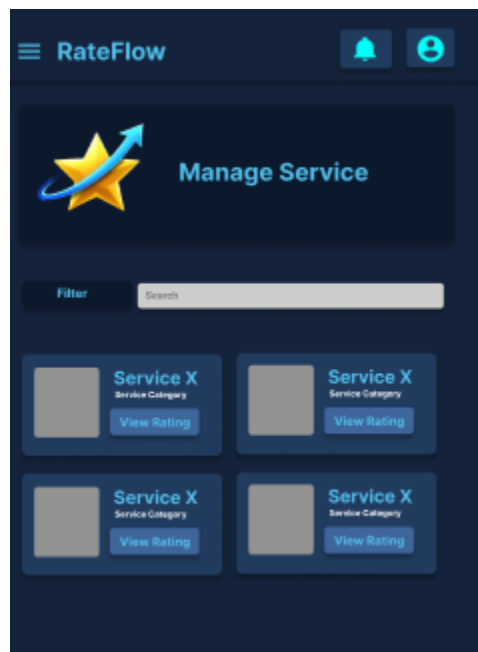
Top Navbar + Scrollable Content

Centered Form Card

Dark-themed UI

Mobile-optimized spacing and touch-friendly controls

Mobile Admin – Manage Service Page



Header:

Menu Icon | App Name ("RateFlow") | Notification Icon | User Profile Icon

Content (Scrollable Single Column):

Banner Section:

App Logo

Title: "Manage Service"

Filter Section:

Filter Button

Search Bar

Services Grid (2 Columns Mobile Layout):**Each Service Card:**

Service Image / Thumbnail

Service Name

Service Category

"View Rating" Button

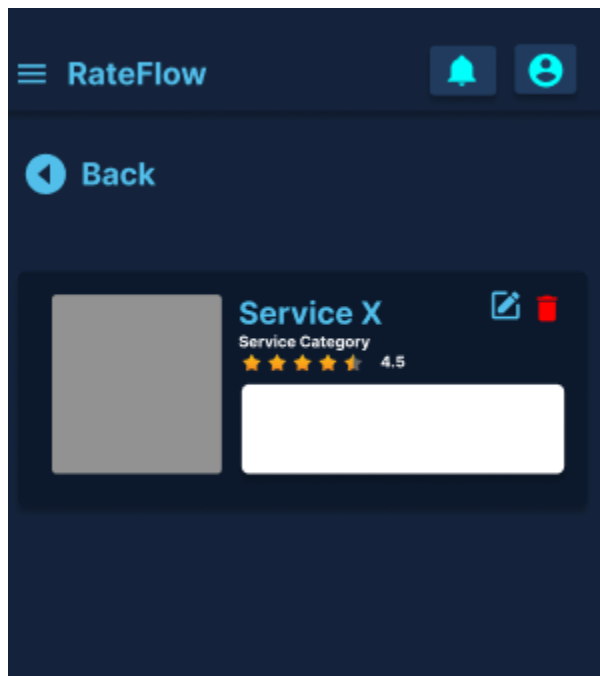
Layout:

Top Navbar + Scrollable Content

2-Column Card Grid

Dark-themed UI

Mobile-optimized spacing and touch-friendly controls

Mobile Admin – Edit Service Page**Header:**

Menu Icon | App Name ("RateFlow") | Notification Icon | User Profile Icon

Back Button (Below Header)

Content (Single Column Layout):

Service Details Card:

Service Image / Thumbnail

Service Name

Service Category

Average Rating Display (Stars + Value)

Service Description Box

Actions:

Edit Icon

Delete Icon

Layout:

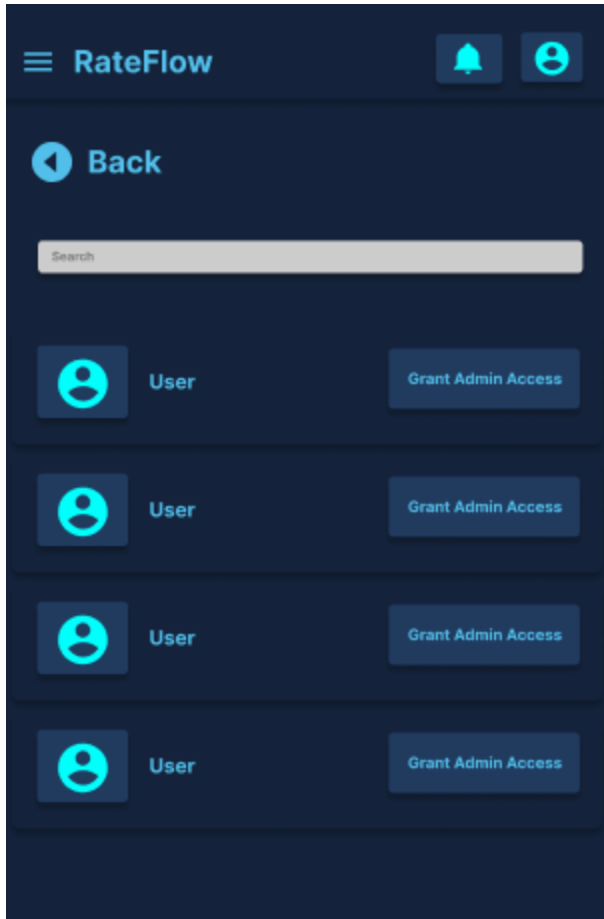
Top Navbar + Scrollable Content

Single Column Layout

Dark-themed UI

Mobile-optimized spacing and touch-friendly controls

Mobile Admin – Access Control Page



Header:

Menu Icon | App Name ("RateFlow") | Notification Icon | User Profile Icon

Back Button (Below Header)

Content (Scrollable Single Column):

Search Bar

Users List (Vertical Layout):

Each User Card:

User Icon

Username

"Grant Admin Access" Button

Layout:

Top Navbar + Scrollable Content

Stacked User Cards

Dark-themed UI
Mobile-optimized spacing and touch-friendly controls

Design System

Colors:

Primary: #5BBAD5 (Main buttons, highlights)
Secondary: #1F3A5F (Cards, elevated surfaces)
Background: #0F1C2E (Main dark background)
Success: #22C55E (Success states, confirmations)
Error: #EF4444 (Delete, destructive actions)
Text Primary: #E6F0FF
Text Secondary: #9FB3C8

Typography:

Font Family: Inter, sans-serif
Headings: Bold, responsive sizing (H1–H4 scale)
Body Text: Regular / Medium
Buttons: Medium weight, uppercase optional

Spacing:

8px grid system
Spacing scale: 8px, 16px, 24px, 32px, 40px, 48px
Consistent padding inside cards and sections

Components:

Buttons: Primary, Secondary, Danger, Disabled states
Inputs: Rounded, dark background, focus highlight
Cards: Soft shadow, rounded corners
Modals: Centered overlay, dimmed background
Icons: Consistent stroke style

Responsive:

Mobile-first approach

Breakpoints:

- 640px (Small tablets)
- 768px (Tablet)
- 1024px (Desktop)
Grid adapts: 1 column (mobile), 2 columns (tablet), 3 columns (desktop)

8.0 PLAN

8.1 Project Timeline

Phase 1: Planning & Design (Week 1-2)

Week 1: Requirements & Architecture

Day 1-2: Project setup and documentation

Day 3-4: Complete FRS and NFR

Day 5-7: System architecture design

Week 2: Detailed Design

Day 1-2: API specification

Day 3-4: Database design

Day 5-6: UI/UX wireframes

Day 7: Implementation plan finalization

Phase 2: Backend Development (Week 3-4)

Week 3: Foundation

Day 1: Spring Boot setup with dependencies

Day 2: Database configuration and entities
Day 3: JWT authentication implementation
Day 4: User management endpoints
Day 5: Product CRUD operations

Week 4: Core Features

Day 1: Cart functionality
Day 2: Order management
Day 3: Search and filtering
Day 4: Error handling and validation
Day 5: API documentation and testing

Phase 3: Web Application (Week 5-6)

Week 5: Frontend Foundation

Day 1: React setup with TypeScript
Day 2: Authentication pages (login, register)
Day 3: Product listing page
Day 4: Product detail page
Day 5: Shopping cart implementation

Week 6: Complete Web Features

Day 1: Checkout flow
Day 2: Order history and confirmation
Day 3: Admin dashboard
Day 4: Responsive design polish
Day 5: API integration and testing

Phase 4: Mobile Application (Week 7-8)

Week 7: Android Foundation

Day 1: Android Studio setup and project structure

Day 2: Authentication screens

Day 3: Product browsing

Day 4: Shopping cart

Day 5: API service layer

Week 8: Complete Mobile App

Day 1: Checkout flow

Day 2: Order management

Day 3: UI polish and animations

Day 4: Testing on emulator/device

Day 5: APK generation and documentation

Phase 5: Integration & Deployment (Week 9-10)

Week 9: Integration Testing

Day 1: End-to-end testing across platforms

Day 2: Bug fixes and optimization

Day 3: Security review

Day 4: Performance testing

Day 5: Documentation updates

Week 10: Deployment

Day 1: Backend deployment (Railway/Heroku)

Day 2: Web app deployment (Vercel/Netlify)

Day 3: Mobile APK distribution

Day 4: Final testing

Day 5: Project submission

Milestones:

- **M1 (End Week 2):** All design documents complete
- **M2 (End Week 4):** Backend API fully functional
- **M3 (End Week 6):** Web application complete
- **M4 (End Week 8):** Mobile application complete
- **M5 (End Week 10):** Full system deployed and integrated

Critical Path:

1. Authentication system (Week 3)
2. Product catalog API (Week 3-4)
3. Shopping cart functionality (Week 4)
4. Checkout process (Week 6)
5. Cross-platform testing (Week 9)

Risk Mitigation:

- Start with simplest working version of each feature
- Test integration points early and often
- Keep backup of working versions
- Focus on core functionality before enhancements